THIRSTY'S CODEX

STANDARDS EDITION

VOLUME XI / 11 OF 20

SUPPORT, ENTERPRISE, AND ROADMAPS

Lifecycle Sections 333-373

NO PROOF, NO CLAIM

CONTINUOUS MASTER FOLIO / SOURCE-PRESERVING EDITION

Edition Statement

THIRSTY'S CODEX / STANDARDS EDITION / SOURCE-PRESERVING SET

This volume is one part of a continuous 20-volume edition. Canonical lifecycle section numbers remain stable across the set; no volume restarts or renumbers them.

The visual treatment has been rebuilt as a unified standards library: coordinated covers, volume architecture, source dividers, running headers, continuous master folios, local page numbers, and source-identity marks.

The source corpus remains authoritative over the presentation layer. Status terms are preserved exactly. CONTENT-COMPLETE does not silently become source-reconciled, hostile-reviewed, owner-accepted, ratified, or published.

Every registered source witness is identified by path, byte count, and SHA-256 digest. The complete source archive is preserved as an embedded attachment in Volume XX, while the two supplied foundational PDFs are preserved as attachments in Volume I.

CONTINUITY RULE

Read the master folio and canonical section sequence as one uninterrupted work. Each succeeding volume begins where the preceding volume ends.

Standards Edition Set Map

VOLUME I
Foundations and Governance
Doctrine, prompts, front matter, shared governance, roles, and output contracts

VOLUME II
Research, Strategy, and Risk
Lifecycle Sections 001-018

VOLUME III
Requirements through Hardware Architecture
Lifecycle Sections 019-060

VOLUME IV
Electrical Engineering through User Validation
Lifecycle Sections 061-106

VOLUME V
Software Systems through Smart Contracts
Lifecycle Sections 107-150

VOLUME VI
Data, Intelligence, and Human Oversight
Lifecycle Sections 151-188

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Cybersecurity through Privacy Engineering
Lifecycle Sections 189-218

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Law, Assurance, and Validation
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VOLUME IX
Configuration through Delivery Infrastructure
Lifecycle Sections 258-300

VOLUME X
Production Operations and Command
Lifecycle Sections 301-332

VOLUME XI
Support, Enterprise, and Roadmaps
Lifecycle Sections 333-373

VOLUME XII
Maintenance, Decommissioning, and Retirement
Lifecycle Sections 374-406

VOLUME XIII
Reference Registers and Working Documentation
Indexes, cross-references, source registers, checkpoints, and operational guides

VOLUME XIV
Standards Source Library
Preserved standards, command editions, blueprints, and source artifacts

VOLUME XV
Canonical Authority and Assurance
Authority, constitutions, policies, schemas, provenance, evaluation, and verifica...

VOLUME XVI
Applied Corpus and V3Q
Agent-output corpus, V3Q implementation, fixtures, tests, and governed examples

VOLUME XVII
Training Views and Indexes
Continuous-pretraining, evaluation, retrieval, and training views with canonical ...

VOLUME XVIII
Agents and Owner Command Standards
Agents 101 and Owner's Cockpit source-standard facsimiles

VOLUME XIX
Static Analysis and Tooling Evidence
Codacy configuration, rulesets, analysis logs, and repository tooling

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333

SECTION 333 - TECHNICAL SUPPORT

CODEX/03_SECTIONS/333_TECHNICAL_SUPPORT.txt

SHA-256 ab26546fe5553f6601b9a84b2ba48673183d995cc15d461c6f5e2518e775c641

4,917 BYTES / PRESERVED SOURCE IDENTITY

SECTION 333 — Technical Support

STATUS: CONTENT-COMPLETE

PURPOSE

Help users and operators diagnose, resolve, work around, and recover from technical problems while preserving security, privacy, evidence, accessibility, and learning.

REQUIRED OUTCOME

A support model with intake, identity, severity, context, troubleshooting, escalation, communication, workaround, resolution, knowledge, privacy, and feedback ownership.

WHY THIS MATTERS

Support is part of the product's real behavior. Poor support can expose data, repeat unsafe advice, blame users, hide defects, and leave people without an accessible recovery path.

APPLICABILITY

Apply when a person or team needs technical assistance to use, operate, integrate, recover, or understand a capability.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no user or operator support path, technical issue, commitment, or recovery need changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use service management, application and platform operations, privacy, security, identity, knowledge, accessibility, incident, and support baselines.

REQUIRED INPUTS

User, entitlement, issue, environment, version, steps, evidence, severity, impact, privacy, workaround, owner, escalation, and resolution criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Support owners protect users, verify access, triage, escalate, communicate, and record. Humans approve sensitive diagnosis, account or data action, compensation, and closure.

AI RESPONSIBILITIES AND BOUNDARIES

AI may classify tickets, suggest known steps, summarize context, and draft responses from approved knowledge. It must not request secrets, expose private data, change user state, make unsupported claims, or close high-impact cases without human review.

APPLICABLE EXISTING AGENT ROLES

Technical Support Engineer, Support Reviewer, Application Engineer, Systems Administrator, Security Auditor, User Experience Reviewer, Analyst, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Intake and authenticate; capture context; classify; troubleshoot safely; provide workaround; escalate; communicate; resolve; verify user outcome; document; identify defect; and improve knowledge.

HUMAN-AI INTERACTION REQUIREMENTS

Show user, issue, version, evidence, action, source, privacy, confidence, and limitation. Require human review for sensitive, security, account, data, safety, and closure decisions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Provide ticket identity, context, redaction, knowledge links, escalation, diagnostic access, audit, user-visible status, and secure support tools with least privilege.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show queue, severity, owner, user impact, evidence freshness, escalation, SLA, privacy, and authorized assign, escalate, communicate, resolve, reopen, and close controls.

REQUIRED OUTPUTS AND ARTIFACTS

Support charter; intake and severity; troubleshooting and escalation; response and workaround; resolution and user confirmation; knowledge, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify user, entitlement, issue, version, environment, source, method, time, action, result, and limitation. Ticket closure proves workflow state, not user success.

VERIFICATION AND ACCEPTANCE TESTS

Verify authentication, privacy, diagnosis, safe advice, escalation, accessibility, communication, workaround, resolution, reopen, and knowledge capture. Test missing context and unavailable support tools.

FAILURE STATES AND ANTI-PATTERNS

Secrets requested, copy-paste diagnosis, blame, unsupported workaround, privacy exposure, no escalation, inaccessible channel, and closure without user outcome.

RECOVERY AND RECONSIDERATION

Protect user and data, correct advice, escalate, preserve evidence, communicate, restore or compensate, and reassess. Reopen on repeated issue, new evidence, security concern, or user dispute.

COMPLETION CRITERIA

Users, issue, severity, evidence, authority, response, escalation, resolution, privacy, accessibility, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve tickets, evidence, responses, workarounds, incidents, decisions, knowledge, exceptions, and open gaps. Hand off to Customer Support, Help Desk, Engineering, Security, Service Management, and Operations.

RELATED CODEX SECTIONS

311 Service Management; 312 IT Operations; 335 Help Desk; 341 Documentation; 342 Knowledge Management; 301-310 Operations; 319 Incident Management.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, service, user, privacy, security, support, and operations context, and the integration roadmap. Exact users, channels, staffing, SLAs, privacy, escalation, and acceptance authority remain unresolved.

334

SECTION 334 - CUSTOMER SUPPORT

CODEX/03_SECTIONS/334_CUSTOMER_SUPPORT.txt

SHA-256 d0d4104e2d67ce4cec54920a287c17c0636929c27384062f0465ced32673d2ab

4,901 BYTES / PRESERVED SOURCE IDENTITY

SECTION 334 — Customer Support

STATUS: CONTENT-COMPLETE

PURPOSE

Provide accessible, respectful, secure, and accountable assistance that helps customers achieve intended outcomes, recover from failure, and influence product improvement.

REQUIRED OUTCOME

A customer-support model with customer identity, entitlement, channel, issue, priority, communication, resolution, escalation, privacy, accessibility, service level, and feedback.

WHY THIS MATTERS

Customers experience the whole service, including support. A fast but inaccurate answer, inaccessible channel, or unowned escalation can increase harm and erode trust.

APPLICABILITY

Apply when customers need help using, integrating, billing, securing, recovering, or understanding a product or service.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no customer, support, entitlement, service commitment, or recovery path changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use service management, technical support, customer success, privacy, security, billing, legal, accessibility, knowledge, and incident baselines.

REQUIRED INPUTS

Customer, entitlement, issue, context, version, evidence, priority, impact, channel, service level, privacy, escalation, resolution, and feedback.

HUMAN RESPONSIBILITIES AND AUTHORITY

Support owners protect customer data, set priority, communicate, escalate, and verify outcome. Humans approve refunds, access, security, legal, safety, and high-impact closure.

AI RESPONSIBILITIES AND BOUNDARIES

AI may classify and summarize authorized cases and draft approved responses. It must not impersonate authority, request secrets, expose data, promise remedy, change account or billing, or close disputed cases without human review.

APPLICABLE EXISTING AGENT ROLES

Customer Support Specialist, Technical Support Engineer, Customer Success Manager, User Experience Reviewer, Privacy Reviewer, Security Auditor, Analyst, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Receive; authenticate; understand outcome; classify; respond; troubleshoot; escalate; communicate; resolve; verify; support appeal; record feedback; and improve product or knowledge.

HUMAN-AI INTERACTION REQUIREMENTS

Show customer, entitlement, issue, evidence, source, action, impact, privacy, and limitation. Require human review for account, billing, security, legal, safety, compensation, and disputed decisions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Provide secure case identity, entitlement, redaction, consent, knowledge, routing, escalation, status, accessibility, audit, and customer-visible history.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show queue, customer impact, priority, owner, SLA, evidence freshness, escalation, sentiment or feedback without treating it as truth, and authorized assign, communicate, resolve, reopen, and compensate controls.

REQUIRED OUTPUTS AND ARTIFACTS

Customer-support charter; channel and entitlement; case and resolution; escalation and accessibility; feedback, incident, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify customer, entitlement, case, source, version, method, time, action, result, and limitation. Customer rating or closure proves only that interaction and response.

VERIFICATION AND ACCEPTANCE TESTS

Verify identity, privacy, channel, accessibility, response, escalation, knowledge, resolution, reopen, communication, and feedback capture. Test vulnerable customer and unavailable channel.

FAILURE STATES AND ANTI-PATTERNS

Script over truth, customer data exposure, inaccessible support, no escalation, unsupported promise, metric gaming, and customer complaint dismissed as noise.

RECOVERY AND RECONSIDERATION

Protect customer, correct response, escalate, communicate, remediate, and reassess harm and commitment. Reopen on dispute, repeated failure, new evidence, or product change.

COMPLETION CRITERIA

Customers, outcomes, channels, entitlement, support, privacy, accessibility, evidence, owners, escalation, resolution, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve cases, responses, escalations, feedback, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Technical Support, Help Desk, Customer Success, Product, Legal, Security, and Operations.

RELATED CODEX SECTIONS

311 Service Management; 333 Technical Support; 335 Help Desk; 336 Customer Success; 341-342 Documentation and Knowledge; 301-310 Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, service, customer, privacy, support, and operations context, and the integration roadmap. Exact customers, channels, SLAs, staffing, data, accessibility, escalation, and acceptance authority remain unresolved.

335

SECTION 335 - HELP DESK

CODEX/03_SECTIONS/335_HELP_DESK.txt

SHA-256 ca3c373ea5109e0bfe7c4205f92c31d590e4dda91526003baebc4ea6a1cbbb2d

4,559 BYTES / PRESERVED SOURCE IDENTITY

SECTION 335 — Help Desk

STATUS: CONTENT-COMPLETE

PURPOSE

Provide the accessible first point of contact for requests, incidents, access, devices, applications, information, and routing to specialized support.

REQUIRED OUTCOME

A help-desk model with intake, identity, categorization, priority, knowledge, communication, escalation, service level, privacy, accessibility, and closure verification.

WHY THIS MATTERS

The help desk shapes the first interpretation of a problem and can either restore work or send users through unsafe loops. A closed ticket is not proof that the user can work.

APPLICABILITY

Apply when users need a central channel for IT, service, access, device, application, or operational assistance.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no central request, incident, access, routing, or user-support workflow changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use IT operations, service management, IAM, technical support, knowledge, privacy, security, accessibility, and incident baselines.

REQUIRED INPUTS

User, identity, request, asset, issue, evidence, category, priority, knowledge, service level, escalation, and resolution.

HUMAN RESPONSIBILITIES AND AUTHORITY

Help-desk owners authenticate, protect data, route, communicate, and verify outcome. Humans approve access changes, device action, sensitive support, and disputed closure.

AI RESPONSIBILITIES AND BOUNDARIES

AI may classify requests, suggest approved knowledge, and draft responses. It must not request secrets, grant access, wipe devices, expose records, or close high-impact work without human review.

APPLICABLE EXISTING AGENT ROLES

Help Desk Analyst, IT Operations Engineer, Technical Support Engineer, IAM Engineer, User Experience Reviewer, Privacy Reviewer, Analyst, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Intake; authenticate; categorize; prioritize; use knowledge; resolve or route; communicate; escalate; verify; record; identify recurring problems; and improve.

HUMAN-AI INTERACTION REQUIREMENTS

Show user, request, identity, asset, category, action, evidence, privacy, and limitation. Require human approval for access, device, security, data, and closure actions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use secure case records, identity, knowledge, routing, status, accessibility, audit, least privilege, redaction, and escalation integration.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show queue, category, priority, owner, SLA, escalation, evidence freshness, and authorized assign, escalate, communicate, resolve, reopen, and close controls.

REQUIRED OUTPUTS AND ARTIFACTS

Help-desk charter; intake and taxonomy; knowledge and routing; service-level and escalation; resolution, feedback, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify user, request, asset, identity, source, method, time, action, result, and limitation. Ticket status proves only workflow state.

VERIFICATION AND ACCEPTANCE TESTS

Verify authentication, privacy, accessibility, categorization, routing, response, escalation, resolution, reopen, and user confirmation. Test urgent and inaccessible-channel cases.

FAILURE STATES AND ANTI-PATTERNS

Password or secret collection, ticket closure theater, no priority, inaccessible channel, wrong routing, unowned escalation, and user blamed for system failure.

RECOVERY AND RECONSIDERATION

Protect user and data, reopen, escalate, correct, communicate, and reassess. Reopen on repeated issue, security concern, user dispute, or service change.

COMPLETION CRITERIA

Users, requests, identity, routing, priority, knowledge, privacy, accessibility, evidence, owners, escalation, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve tickets, routing, knowledge, escalations, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Customer Support, Technical Support, IT Operations, IAM, Security, and Service Management.

RELATED CODEX SECTIONS

311-312 Service and IT Operations; 333-334 Technical and Customer Support; 341-342 Documentation and Knowledge; 200-202 IAM; 319 Incident Management.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, support, service, identity, privacy, and operations context, and the integration roadmap. Exact users, channels, staffing, SLAs, identity, accessibility, escalation, and acceptance authority remain unresolved.

336

SECTION 336 - CUSTOMER SUCCESS

CODEX/03_SECTIONS/336_CUSTOMER_SUCCESS.txt

SHA-256 c080b8c666ba626a9778fded34d1388531d38a096eb8b3f87c43c621340f0a91

4,775 BYTES / PRESERVED SOURCE IDENTITY

SECTION 336 — Customer Success

STATUS: CONTENT-COMPLETE

PURPOSE

Help customers achieve their intended outcomes through adoption, enablement, measurement, risk identification, support coordination, and honest feedback.

REQUIRED OUTCOME

A customer-success model with customer goals, success criteria, adoption, health, value evidence, risks, plan, support, escalation, renewal or continuation, and feedback.

WHY THIS MATTERS

Usage or renewal can hide whether customers achieve value, understand risk, or feel able to leave. Success claims must reflect customer outcomes rather than account activity alone.

APPLICABILITY

Apply when an owner or team helps customers adopt, operate, measure, expand, renew, or recover value from a product or service.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no ongoing customer outcome, adoption, value, support, or relationship decision changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use product, user research, service, support, analytics, privacy, security, legal, training, and roadmap baselines.

REQUIRED INPUTS

Customer goals, success criteria, users, use case, adoption, health, outcomes, risks, support, incidents, feedback, renewal, and evidence.

HUMAN RESPONSIBILITIES AND AUTHORITY

Customer-success owners represent customer outcomes, communicate honestly, escalate risk, protect data, and avoid making commitments they cannot authorize. Humans decide remedy, scope, and relationship action.

AI RESPONSIBILITIES AND BOUNDARIES

AI may summarize authorized customer signals, identify risk, draft enablement and review questions, and suggest next steps. It must not infer sentiment or intent as fact, expose data, promise outcome, or make account decisions without human review.

APPLICABLE EXISTING AGENT ROLES

Customer Success Manager, User Researcher, Product Manager, Analyst, Technical Support Engineer, Privacy Reviewer, User Experience Reviewer, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Understand goals; define success; plan adoption; enable; measure outcome; monitor risk; coordinate support; communicate; review value; address gaps; capture feedback; and decide continuation or change.

HUMAN-AI INTERACTION REQUIREMENTS

Show customer, goal, use case, evidence, health, risk, recommendation, privacy, and limitation. Require human review for customer communication, commitment, risk, renewal, and remedy.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Support customer outcome instrumentation, privacy, role separation, accessible enablement, feedback, support escalation, status, and data export or exit.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show customer goals, outcome, adoption, health, risks, incidents, support, evidence freshness, owner, and authorized intervene, escalate, communicate, compensate, and close controls.

REQUIRED OUTPUTS AND ARTIFACTS

Success charter; customer goal and health; adoption and outcome; enablement and support; risk, feedback, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify customer, goal, use case, metric, source, period, method, result, uncertainty, and limitation. Usage, satisfaction, or renewal proves only its bounded signal.

VERIFICATION AND ACCEPTANCE TESTS

Verify goal, success criterion, user outcome, enablement, accessibility, support, risk, privacy, feedback, and customer understanding. Test failure and recovery journey.

FAILURE STATES AND ANTI-PATTERNS

Adoption as value, renewal as success, hidden churn risk, unsupported promise, customer data overcollection, and success narrative that ignores dissent or harm.

RECOVERY AND RECONSIDERATION

Protect customer, acknowledge mismatch, coordinate support, provide remedy, preserve feedback, and reassess outcome. Reopen on goal, product, user, risk, or relationship change.

COMPLETION CRITERIA

Customer, goals, outcomes, evidence, adoption, risk, support, owner, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending customer and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve success plans, feedback, risks, incidents, support, decisions, evidence, exceptions, and open gaps. Hand off to Customer Support, Product, Sales, Solutions, Training, and Operations.

RELATED CODEX SECTIONS

001 User Research; 010 Product Management; 311 Service Management; 334 Customer Support; 338 Training; 343 Product Marketing; 373 Roadmap Management.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, customer, product, service, support, privacy, and outcome context, and the integration roadmap. Exact customers, goals, measures, data, commitments, staffing, and acceptance authority remain unresolved.

337

SECTION 337 - COMMUNITY MANAGEMENT

CODEX/03_SECTIONS/337_COMMUNITY_MANAGEMENT.txt

SHA-256 8a5f3491a57db8a3dfc481f019bbba065d6bbb148e016abcf326e21e3783a2e4

4,890 BYTES / PRESERVED SOURCE IDENTITY

SECTION 337 — Community Management

STATUS: CONTENT-COMPLETE

PURPOSE

Cultivate and govern a community around a product, domain, mission, or practice through participation, norms, safety, support, learning, feedback, and accountable stewardship.

REQUIRED OUTCOME

A community model with purpose, members, norms, moderation, accessibility, safety, privacy, participation, feedback, escalation, events, evidence, and continuity.

WHY THIS MATTERS

Communities create knowledge and trust but can also concentrate harm, silence dissent, expose people, or burden unpaid contributors. Activity is not belonging, health, or safety.

APPLICABILITY

Apply when people interact through a shared community, forum, event, network, repository, program, or practice with organizational stewardship.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no community, member, norm, moderation, safety, participation, or feedback behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use trust and safety, privacy, accessibility, policy, user research, support, events, legal, ethics, and governance baselines.

REQUIRED INPUTS

Purpose, members, norms, channels, moderators, reports, safety, accessibility, privacy, events, feedback, disputes, evidence, and escalation.

HUMAN RESPONSIBILITIES AND AUTHORITY

Community owners set purpose, norms, moderation, safety, appeals, privacy, and stewardship. Humans review high-impact enforcement, vulnerable-person concerns, and disputes.

AI RESPONSIBILITIES AND BOUNDARIES

AI may summarize authorized discussion, identify themes, draft moderation queues, and suggest resources. It must not expose members, infer intent, remove speech, punish, or represent community consensus without human review.

APPLICABLE EXISTING AGENT ROLES

Community Manager, Trust and Safety Reviewer, User Researcher, Moderator, Privacy Reviewer, Accessibility Reviewer, Analyst, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Define purpose and norms; onboard; moderate; protect privacy and safety; support accessibility; host events; gather feedback; handle conflict; appeal; measure health; and revise stewardship.

HUMAN-AI INTERACTION REQUIREMENTS

Show member, context, norm, report, evidence, action, privacy, and limitation. Require human moderation for removal, sanctions, disclosure, and appeals. Do not treat volume or sentiment as consensus.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Provide reporting, moderation, evidence, privacy, identity, role, appeal, accessibility, status, and safe event controls. Protect volunteer and vulnerable participants.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show community health, reports, moderation, accessibility, participation, incidents, evidence freshness, owner, and authorized protect, restrict, appeal, communicate, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Community charter; membership and norms; moderation and appeal; safety and accessibility; event and feedback; decision, evidence, incident, and continuity records.

EVIDENCE REQUIREMENTS

Identify member or segment, context, report, source, method, time, action, result, impact, and limitation. Activity, sentiment, or report count proves only a bounded signal.

VERIFICATION AND ACCEPTANCE TESTS

Verify onboarding, norms, reporting, moderation, privacy, accessibility, appeal, events, conflict, safety, support, and member understanding. Test abuse and moderator failure.

FAILURE STATES AND ANTI-PATTERNS

Engagement as health, opaque moderation, reporter exposure, no appeal, volunteer exploitation, token participation, consensus theater, and safety policy without enforcement.

RECOVERY AND RECONSIDERATION

Protect members, preserve evidence, pause harmful behavior, reverse unjust action, provide support, communicate, and reassess norms or moderation. Reopen on harm, population, platform, policy, or event change.

COMPLETION CRITERIA

Purpose, members, norms, safety, privacy, accessibility, moderation, appeal, evidence, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending community and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve charter, reports, moderation, appeals, events, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Trust and Safety, Support, Education, Product, Legal, Privacy, and Governance.

RELATED CODEX SECTIONS

001 User Research; 216 Abuse Prevention; 217 Trust and Safety; 218 Privacy; 222 Ethics; 334 Customer Support; 338-340 Training, Education, and Developer Relations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, community, trust, safety, privacy, and user context, and the integration roadmap. Exact community, members, norms, moderation, accessibility, legal, and acceptance authority remain unresolved.

338

SECTION 338 - TRAINING

CODEX/03_SECTIONS/338_TRAINING.txt

SHA-256 4e69e17dfbca74f48d406a12ee809685626caff0e5b08140fd7c2df7546a095b

4,837 BYTES / PRESERVED SOURCE IDENTITY

SECTION 338 — Training

STATUS: CONTENT-COMPLETE

PURPOSE

Develop and verify the knowledge, skills, judgment, practice, accessibility, and support people need to perform a defined role or use a capability safely and effectively.

REQUIRED OUTCOME

A training model with audience, objectives, prerequisites, content, practice, assessment, accessibility, records, refresh, support, transfer, and outcome evidence.

WHY THIS MATTERS

Completion does not prove competence or safe performance. Training can also create harm when it teaches outdated, inaccessible, unsafe, or context-free procedures.

APPLICABILITY

Apply when people need structured learning to perform, operate, support, govern, or recover a system, process, role, or capability.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no skill, role, safety, compliance, operational, or user outcome changes and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use user research, role, requirements, safety, security, privacy, accessibility, documentation, operations, and legal baselines.

REQUIRED INPUTS

Audience, role, objectives, prerequisites, content, practice, assessment, environment, accessibility, instructor, records, refresh, and success measure.

HUMAN RESPONSIBILITIES AND AUTHORITY

Training owners define competence and safety; subject experts validate content; humans assess learners and approve records. Learners can ask, fail safely, and request accommodation.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft exercises, explain approved material, personalize practice, and summarize feedback. It must not certify competence, invent policy, expose learner data, or replace instructor or assessor judgment.

APPLICABLE EXISTING AGENT ROLES

Training Designer, Teacher, Subject Matter Expert, User Experience Reviewer, Accessibility Reviewer, Researcher, Tester, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Define outcome and audience; assess prerequisite; design content and practice; deliver accessibly; assess; record; support; refresh; measure transfer; and retire obsolete training.

HUMAN-AI INTERACTION REQUIREMENTS

Show objective, source, learner, exercise, assessment, evidence, privacy, and limitation. Require human review for safety, policy, certification, and accommodation decisions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Provide versioned content, sandbox practice, safe failure, identity and records, accessibility, privacy, assessment integrity, and support for offline or alternate formats.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show course, learner group, objective, completion, assessment, accessibility, content version, evidence freshness, owner, and authorized revise, accommodate, recertify, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Training charter; audience and competency; content and practice; assessment and accommodation; completion and transfer; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify learner or group, objective, content version, activity, assessment, method, time, result, assessor, and limitation. Completion proves participation, not competence or transfer.

VERIFICATION AND ACCEPTANCE TESTS

Verify content accuracy, accessibility, practice, assessment, safety, privacy, records, learner understanding, transfer, refresh, and support. Test outdated content and accommodation.

FAILURE STATES AND ANTI-PATTERNS

Completion as competence, inaccessible course, outdated content, unsafe practice, assessment leakage, no refresh, and training used to transfer system responsibility to users.

RECOVERY AND RECONSIDERATION

Correct content, notify learners, suspend unsafe certification, provide support, reassess competence, and update controls. Reopen on role, policy, system, safety, or content change.

COMPLETION CRITERIA

Audience, objectives, content, practice, assessment, accessibility, evidence, owners, refresh, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending subject-matter and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve content versions, assessments, accommodations, records, feedback, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Education, Documentation, Support, Operations, Security, and Governance.

RELATED CODEX SECTIONS

001 User Research; 007 Glossary; 048 Accessibility; 338 Training; 339 Education; 341 Documentation; 331 Operational Readiness; 402-404 Review.

CODEX/03_SECTIONS/338_TRAINING.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, user, role, safety, accessibility, documentation, and operations context, and the integration roadmap. Exact audience, competence, content, assessment, records, accommodation, and acceptance authority remain unresolved.

339

SECTION 339 - EDUCATION

CODEX/03_SECTIONS/339_EDUCATION.txt

SHA-256 8c021f2cdebd8d2f525d89d145247295649cd845eb22ad29da2fe5960761c784

4,666 BYTES / PRESERVED SOURCE IDENTITY

SECTION 339 — Education

STATUS: CONTENT-COMPLETE

PURPOSE

Build durable understanding, judgment, and shared language about a domain, system, practice, or capability beyond one immediate task.

REQUIRED OUTCOME

An education model with audience, learning purpose, concepts, sources, progression, inquiry, practice, accessibility, assessment, feedback, and continuity.

WHY THIS MATTERS

Education shapes future decisions and agency. Simplified material can become misleading doctrine, while inaccessible or one-sided instruction can reproduce blind spots.

APPLICABILITY

Apply when the goal is durable conceptual understanding, professional development, public literacy, community capability, or shared practice.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no learning, shared understanding, public or professional capability, or decision quality outcome changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use research, domain analysis, ethics, accessibility, documentation, training, legal, source, and knowledge baselines.

REQUIRED INPUTS

Audience, goals, concepts, source basis, perspectives, sequence, examples, practice, assessment, accessibility, instructor, feedback, and limitations.

HUMAN RESPONSIBILITIES AND AUTHORITY

Educators and subject experts choose sources, represent uncertainty, protect learners, and assess understanding. Humans decide what is contested, required, or suitable for a population.

AI RESPONSIBILITIES AND BOUNDARIES

AI may explain approved concepts, generate examples, compare perspectives, and adapt practice. It must not fabricate sources, erase disagreement, certify understanding, or teach unsafe or contested claims as settled fact.

APPLICABLE EXISTING AGENT ROLES

Teacher, Researcher, Domain Analyst, Documentation Writer, User Experience Reviewer, Accessibility Reviewer, Critic, Verifier, and Memory Curator.

REQUIRED ACTIVITIES

Define learners and goals; inventory sources; structure concepts; present examples and uncertainty; enable inquiry and practice; assess; collect feedback; update; and preserve continuity.

HUMAN-AI INTERACTION REQUIREMENTS

Show source, perspective, certainty, learner, exercise, feedback, and limitation. Require human review for source, safety, rights, cultural context, and assessment.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use versioned content, source links, accessible formats, learner control, safe practice, privacy, feedback, assessment integrity, and content retirement.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show curriculum, source freshness, perspectives, learner progress, accessibility, evidence, owner, and authorized revise, annotate, challenge, accommodate, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Education charter; audience and learning map; source and concept structure; examples and assessment; feedback, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify claim, source, version, perspective, learner, method, time, result, and limitation. Completion or quiz score proves only bounded participation and recall.

VERIFICATION AND ACCEPTANCE TESTS

Verify source, accuracy, uncertainty, accessibility, comprehension, perspective, safety, feedback, assessment, and content update. Test outdated or conflicting source.

FAILURE STATES AND ANTI-PATTERNS

Source-free lesson, false certainty, one perspective, inaccessible material, assessment as understanding, and education that hides its own limits.

RECOVERY AND RECONSIDERATION

Correct content, notify learners, preserve version, provide clarification, and reassess affected understanding. Reopen on source, domain, audience, policy, or risk change.

COMPLETION CRITERIA

Audience, goals, concepts, sources, perspectives, accessibility, evidence, assessment, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending subject-matter and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve curriculum, sources, versions, learner feedback, decisions, evidence, exceptions, and open gaps. Hand off to Training, Documentation, Knowledge, Community, Product, and Governance.

RELATED CODEX SECTIONS

001 Research; 007 Glossary; 338 Training; 341 Documentation; 342 Knowledge Management; 337 Community; 222 Ethics; 247 Accessibility Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, research, education, documentation, accessibility, and knowledge context, and the integration roadmap. Exact learners, sources, perspectives, standards, assessment, and acceptance authority remain unresolved.

340

SECTION 340 - DEVELOPER RELATIONS

CODEX/03_SECTIONS/340_DEVELOPER_RELATIONS.txt

SHA-256 80d043f85784c930d476a88249699a6923fec6c39658430cae62e3deafdb1a42

4,988 BYTES / PRESERVED SOURCE IDENTITY

SECTION 340 — Developer Relations

STATUS: CONTENT-COMPLETE

PURPOSE

Build two-way relationships with developers and technical communities through enablement, feedback, events, documentation, examples, support, and transparent stewardship.

REQUIRED OUTCOME

A developer-relations model with audience, goals, channels, programs, content, community, feedback, safety, accessibility, evidence, commitments, and handoff.

WHY THIS MATTERS

Developer relations can create trust and learning or become marketing theater. Reach, attendance, and sentiment do not prove technical usefulness, truthful claims, or community health.

APPLICABILITY

Apply when an organization engages developers, technical users, contributors, partners, or communities around a product, platform, standard, or practice.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no developer audience, engagement, enablement, feedback, event, support, or public technical claim changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use product, developer experience, community, documentation, education, support, privacy, legal, IP, marketing, and governance baselines.

REQUIRED INPUTS

Audience, goals, product, technical content, channels, events, community, feedback, support, accessibility, privacy, commitments, and evidence.

HUMAN RESPONSIBILITIES AND AUTHORITY

Developer-relations owners represent product truth, protect community, moderate, communicate limits, and route feedback. Humans approve technical commitments, public claims, sponsorship, and sensitive responses.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft examples, summarize authorized feedback, personalize content, and identify recurring questions. It must not invent capabilities, impersonate staff, expose community data, make commitments, or moderate high-impact issues without human review.

APPLICABLE EXISTING AGENT ROLES

Developer Relations Lead, Developer Experience Engineer, Community Manager, Technical Writer, Product Manager, Researcher, Support Reviewer, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Understand audience; define program; create and review content; host events; support community; collect feedback; communicate roadmap and limits; measure technical outcome; escalate risk; and preserve continuity.

HUMAN-AI INTERACTION REQUIREMENTS

Show audience, source, product version, claim, feedback, privacy, evidence, and limitation. Require human review for public technical claims, moderation, commitments, and safety.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Support feedback capture, content versioning, community safety, accessible events, support routing, consent, analytics, moderation, and durable knowledge transfer.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show audience, program, content, questions, feedback, incidents, evidence freshness, owner, and authorized revise, clarify, escalate, moderate, communicate, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Developer-relations charter; audience and channel; content and event; community and support; feedback, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify audience, claim, product version, source, channel, event, method, time, result, and limitation. Reach, attendance, or social sentiment proves only its bounded signal.

VERIFICATION AND ACCEPTANCE TESTS

Verify technical accuracy, examples, accessibility, community safety, privacy, support routing, feedback, claim limits, and handoff. Test outdated content and difficult question escalation.

FAILURE STATES AND ANTI-PATTERNS

Marketing disguised as technical guidance, invented roadmap, community data exposure, no moderation, engagement as usefulness, and feedback collected without ownership or response.

RECOVERY AND RECONSIDERATION

Correct public material, notify audience, protect participants, preserve feedback, and reassess claim or program. Reopen on product, audience, community, policy, or safety change.

COMPLETION CRITERIA

Audience, goals, content, channels, feedback, safety, accessibility, evidence, owners, commitments, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve content, events, feedback, questions, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Product, Documentation, Education, Community, Support, Marketing, and Governance.

RELATED CODEX SECTIONS

273 Developer Experience; 337 Community Management; 338-339 Training and Education; 341 Documentation; 343-344 Product and Technical Marketing; 373 Roadmap Management.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, developer, community, product, content, support, and marketing context, and the integration roadmap. Exact audience, channels, claims, staffing, privacy, moderation, and acceptance authority remain unresolved.

341

SECTION 341 - DOCUMENTATION ENGINEERING

CODEX/03_SECTIONS/341_DOCUMENTATION_ENGINEERING.txt

SHA-256 dbfa9022457f2276281dc9e578d24f79b3d0ccea81070af7ac7f182eb264af1e

4,894 BYTES / PRESERVED SOURCE IDENTITY

SECTION 341 — Documentation Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Engineer documentation as a versioned, tested, accessible, source-grounded product that helps its audience understand, decide, operate, recover, and maintain continuity.

REQUIRED OUTCOME

A documentation model with audience, purpose, source, structure, terminology, examples, version, accessibility, review, publication, feedback, retirement, and evidence.

WHY THIS MATTERS

Documentation is an operational and governance interface. Stale, ambiguous, inaccessible, or untested instructions can create defects, unsafe action, support load, and false expectations.

APPLICABILITY

Apply when instructions, reference, explanation, decision records, policies, runbooks, training, or public technical content affects people or systems.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no user, operator, decision, support, governance, or continuity documentation is created or changed and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use user research, glossary, source, product, architecture, operations, accessibility, localization, legal, IP, and knowledge baselines.

REQUIRED INPUTS

Audience, purpose, source, version, terms, task, procedure, examples, limits, accessibility, localization, review, publication, and feedback.

HUMAN RESPONSIBILITIES AND AUTHORITY

Documentation owners and subject experts verify accuracy, scope, safety, and approval. Humans decide normative language, public claim, legal text, and retirement.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft, restructure, summarize supplied sources, identify gaps, and propose examples. It must not invent facts, citations, behavior, policy, or user evidence, and must not publish authoritative material without human review.

APPLICABLE EXISTING AGENT ROLES

Documentation Writer, Technical Writer, Researcher, Subject Matter Expert, User Experience Reviewer, Accessibility Reviewer, Verifier, Critic, and Memory Curator.

REQUIRED ACTIVITIES

Define audience and task; gather authoritative sources; write; review; test instructions and examples; validate accessibility and localization; publish; collect feedback; update; version; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show source, version, audience, claim, uncertainty, example, reviewer, and limitation. Require human review for accuracy, safety, legal, public, operational, and normative content.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use docs-as-code or equivalent versioning, source links, metadata, search, accessibility, localization, publication gates, redirects, feedback, and archival.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show document status, source freshness, audience, owner, review, feedback, evidence, accessibility, and authorized edit, approve, publish, correct, deprecate, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Documentation charter; audience and content map; source and glossary; drafts and examples; review and publication; feedback, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify claim, source, version, audience, task, method, reviewer, time, result, and limitation. Publication or search ranking proves neither correctness nor usefulness.

VERIFICATION AND ACCEPTANCE TESTS

Verify source, procedure, examples, links, version, accessibility, localization, search, task success, safety, support, and retirement. Test clean reader and failure path.

FAILURE STATES AND ANTI-PATTERNS

Stale docs, invented example, citation without source, inaccessible format, no owner, duplicate authority, broken link, and documentation that hides uncertainty or recovery.

RECOVERY AND RECONSIDERATION

Correct or withdraw, notify readers, preserve previous version, provide support, and reassess affected action. Reopen on product, policy, source, user, locale, or operational change.

COMPLETION CRITERIA

Audience, purpose, source, version, content, review, accessibility, publication, evidence, owner, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending subject-matter and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve source, versions, reviews, links, feedback, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Knowledge, Training, Support, Developer Relations, Product, and Operations.

RELATED CODEX SECTIONS

007 Glossary; 001 Research; 273 Developer Experience; 338-340 Training, Education, and Developer Relations; 342 Knowledge; 388 Version Management.

CODEX/03_SECTIONS/341_DOCUMENTATION_ENGINEERING.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, documentation, source, user, accessibility, operations, and continuity context, and the integration roadmap. Exact audiences, sources, owners, review, localization, publication, and acceptance authority remain unresolved.

342

SECTION 342 - KNOWLEDGE MANAGEMENT

CODEX/03_SECTIONS/342_KNOWLEDGE_MANAGEMENT.txt

SHA-256 68858f7e842bbfe750aff0bddfcd4acc3e6b170415cc4d81dc467d2b6ec835db

4,741 BYTES / PRESERVED SOURCE IDENTITY

SECTION 342 — Knowledge Management

STATUS: CONTENT-COMPLETE

PURPOSE

Create, organize, validate, retrieve, share, preserve, and retire knowledge needed for decisions, work, support, learning, governance, and continuity.

REQUIRED OUTCOME

A knowledge model with sources, authority, topics, owners, version, context, access, search, freshness, reuse, feedback, retention, and unresolved questions.

WHY THIS MATTERS

Knowledge systems can spread stale, duplicated, decontextualized, or unauthorized claims. Retrieval speed is not truth, and a polished summary can conceal source gaps.

APPLICABILITY

Apply when information is curated for reuse in decisions, operations, support, training, education, community, AI, or organizational continuity.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no reusable knowledge, source, decision, support, or continuity claim changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use research, documentation, records, data, privacy, security, IP, legal, training, support, and governance baselines.

REQUIRED INPUTS

Sources, claims, topics, authority, owner, version, context, access, audience, freshness, feedback, retention, and unresolved gaps.

HUMAN RESPONSIBILITIES AND AUTHORITY

Knowledge owners curate and verify; subject experts resolve meaning; humans approve authoritative status, access, retention, and retirement. Users can challenge and correct knowledge.

AI RESPONSIBILITIES AND BOUNDARIES

AI may retrieve authorized sources, summarize, classify, identify contradiction, and draft updates. It must preserve provenance and uncertainty and must not elevate, invent, expose, or retire knowledge without human authority.

APPLICABLE EXISTING AGENT ROLES

Knowledge Manager, Researcher, Documentation Writer, Analyst, Domain Expert, Librarian, Verifier, Critic, Privacy Reviewer, and Memory Curator.

REQUIRED ACTIVITIES

Inventory and classify; establish authority; curate and contextualize; index; publish; review freshness; capture feedback and decisions; protect access; archive; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show source, authority, version, context, claim, audience, evidence, uncertainty, and limitation. Require human review for authoritative status, sensitive access, policy, and correction.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use provenance, metadata, search, source links, versioning, access, freshness, contradiction, feedback, retention, archive, and audit. Prevent summaries from replacing source where source matters.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show knowledge item, source, owner, authority, freshness, usage, contradiction, access, evidence, and authorized correct, annotate, restrict, promote, archive, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Knowledge charter; source and authority register; taxonomy and index; content and review; feedback, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify claim, source, version, context, authority, method, time, reviewer, result, and limitation. Search rank, reuse, or summary fluency proves neither truth nor authority.

VERIFICATION AND ACCEPTANCE TESTS

Verify source, provenance, freshness, access, search, context, contradiction, correction, accessibility, retention, and user task. Test stale or conflicting knowledge.

FAILURE STATES AND ANTI-PATTERNS

Knowledge dumping, duplicate authority, stale runbook, source hidden, access leak, AI summary as truth, no correction, and usage metric as usefulness.

RECOVERY AND RECONSIDERATION

Mark stale or unsafe item, correct or withdraw, notify consumers, preserve history, and reassess decisions or operations based on it. Reopen on source, domain, policy, product, or audience change.

COMPLETION CRITERIA

Sources, authority, context, owners, version, access, freshness, evidence, feedback, retention, limits, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve sources, versions, reviews, corrections, usage, decisions, evidence, exceptions, and open gaps. Hand off to Documentation, Training, Support, Education, AI Operations, Records, and Governance.

RELATED CODEX SECTIONS

004 Evidence and Truth; 007 Glossary; 001 Research; 225 Records; 308 AI Operations; 333-342 Support, Training, Documentation, and Knowledge; 391-395 Data Lifecycle.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, knowledge, evidence, source, privacy, and continuity context, and the integration roadmap. Exact sources, authority, taxonomy, freshness, access, retention, and acceptance authority remain unresolved.

343

SECTION 343 - PRODUCT MARKETING

CODEX/03_SECTIONS/343_PRODUCT_MARKETING.txt

SHA-256 be19ef0cb8573ea9600c6e4c4272654b70cf0a5df0d26de3b1f1dc71d4956f73

4,837 BYTES / PRESERVED SOURCE IDENTITY

SECTION 343 — Product Marketing

STATUS: CONTENT-COMPLETE

PURPOSE

Translate a real product capability and customer outcome into truthful positioning, audience understanding, launch, adoption, and feedback without overclaiming.

REQUIRED OUTCOME

A product-marketing model with audience, problem, value, evidence, positioning, claims, alternatives, content, launch, enablement, feedback, and correction.

WHY THIS MATTERS

Marketing shapes expectations and decisions. Claims that outrun product behavior create harm, support load, legal exposure, and loss of trust even when the product itself works.

APPLICABILITY

Apply when a product, service, capability, launch, offer, or customer outcome is positioned, promoted, explained, or compared.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no audience, product claim, launch, adoption, customer expectation, or public content changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use product, user research, market, competitive, legal, IP, privacy, technical, support, accessibility, and roadmap baselines.

REQUIRED INPUTS

Product, audience, problem, outcome, evidence, alternatives, limitations, claims, content, launch, support, feedback, and approval authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Marketing owners ensure claims are truthful and reviewed by product, technical, legal, and accessibility authorities. Humans approve public positioning, comparison, pricing, and correction.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft copy, segment authorized research, summarize evidence, and identify claim gaps. It must not invent capability, fabricate customer proof, conceal limitation, impersonate customer voice, or publish without human approval.

APPLICABLE EXISTING AGENT ROLES

Product Marketer, Product Manager, Researcher, Analyst, Technical Marketer, Legal Advisor, Documentation Writer, Verifier, Critic, and User Experience Reviewer.

REQUIRED ACTIVITIES

Understand audience and problem; define outcome; validate positioning and claims; compare alternatives fairly; create content; coordinate launch and support; monitor feedback; correct overclaim; and retire stale material.

HUMAN-AI INTERACTION REQUIREMENTS

Show claim, source, product version, audience, evidence, limitation, reviewer, and uncertainty. Require human review for legal, technical, customer, competitor, safety, and public claims.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link claims to product behavior, evidence, versions, terms, support, accessibility, pricing, and correction paths. Preserve content version and distribution control.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show claims, evidence, product version, audience, content, feedback, risks, owner, and authorized approve, correct, withdraw, communicate, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Marketing charter; audience and positioning; claim and evidence register; content and launch; enablement, correction, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify claim, product version, source, customer or study, method, date, result, reviewer, and limitation. Reach, testimonial, or conversion proves only the bounded claim.

VERIFICATION AND ACCEPTANCE TESTS

Verify capability, evidence, audience, comparison, legal, accessibility, support, limitation, version, and correction. Test content against actual product and failure path.

FAILURE STATES AND ANTI-PATTERNS

Overclaim, cherry-picked proof, competitor misrepresentation, hidden limitation, synthetic testimonial, old product content, and launch metric treated as value.

RECOVERY AND RECONSIDERATION

Correct or withdraw content, notify affected audiences, support users, preserve evidence, and reassess claims and commitments. Reopen on product, audience, evidence, law, or market change.

COMPLETION CRITERIA

Audience, problem, outcome, claims, evidence, limitations, approvals, content, support, owners, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve claims, sources, versions, approvals, launches, feedback, corrections, decisions, evidence, exceptions, and open gaps. Hand off to Technical Marketing, Sales, Solutions, Product, Legal, Support, and Documentation.

RELATED CODEX SECTIONS

001 User Research; 005 Mission and Strategy; 010 Product Management; 219 Legal; 223 IP; 336 Customer Success; 344 Technical Marketing; 346-347 Sales and Solutions Engineering.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, product, market, customer, legal, technical, and evidence context, and the integration roadmap. Exact audience, claims, proof, reviewers, legal, accessibility, and acceptance authority remain unresolved.

344

SECTION 344 - TECHNICAL MARKETING

CODEX/03_SECTIONS/344_TECHNICAL_MARKETING.txt

SHA-256 fd2196f8743e8cd10a611b058a6eafff88bc253b73075e212841fc6ce4759922

4,753 BYTES / PRESERVED SOURCE IDENTITY

SECTION 344 — Technical Marketing

STATUS: CONTENT-COMPLETE

PURPOSE

Explain technical capability, architecture, integration, performance, security, and operational value accurately to technical audiences and decision makers.

REQUIRED OUTCOME

A technical-marketing model with audience, use case, architecture, claims, evidence, limitations, proof, content, review, enablement, and correction.

WHY THIS MATTERS

Technical audiences act on specifics. Vague or inflated claims about compatibility, security, scale, AI, or operations can cause unsafe design and unrealistic commitments.

APPLICABILITY

Apply when technical features, architecture, benchmarks, security, integration, deployment, or operational claims are communicated externally or internally.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no technical claim, audience, artifact, comparison, or customer expectation changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use architecture, performance, security, testing, product, legal, IP, support, documentation, and release baselines.

REQUIRED INPUTS

Audience, use case, architecture, version, benchmark, security, compatibility, limits, source, content, reviewer, and correction path.

HUMAN RESPONSIBILITIES AND AUTHORITY

Technical and product owners verify claims; legal reviews required content; marketing communicates within approved boundaries. Humans approve benchmark, security, AI, and comparison claims.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft technical content from authorized sources and summarize evidence. It must not fabricate benchmark, customer, security, compatibility, or architecture claims or publish without qualified review.

APPLICABLE EXISTING AGENT ROLES

Technical Marketer, Architect, Product Manager, Engineer, Researcher, Analyst, Legal Advisor, Verifier, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Define audience and use case; gather current technical evidence; draft; review with engineers and users; validate examples and benchmarks; publish; enable sales and support; monitor correction; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show claim, version, source, method, environment, audience, limitation, reviewer, and uncertainty. Require human technical, legal, security, and accessibility review.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link claims to actual architecture, configurations, tests, artifacts, support, and version. Preserve benchmark conditions, terms, diagrams, and correction history.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show technical claim, source, version, evidence freshness, owner, risk, content, feedback, and authorized approve, correct, qualify, withdraw, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Technical-marketing charter; audience and claim register; evidence and benchmark; content and review; enablement, correction, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify claim, version, environment, method, source, result, date, reviewer, and limitation. Benchmark or diagram proves only its declared conditions.

VERIFICATION AND ACCEPTANCE TESTS

Verify technical accuracy, version, benchmark, security, compatibility, architecture, operations, accessibility, legal, support, and correction. Test content against actual behavior.

FAILURE STATES AND ANTI-PATTERNS

Benchmark without workload, security badge as proof, architecture diagram as runtime truth, AI-generated proof, stale content, and technical detail used to hide limitations.

RECOVERY AND RECONSIDERATION

Correct, qualify, or withdraw content, notify users and teams, preserve evidence, and reassess commitments. Reopen on version, benchmark, architecture, threat, or audience change.

COMPLETION CRITERIA

Audience, claims, versions, sources, methods, limitations, reviewers, evidence, content, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending technical and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve claims, benchmarks, diagrams, sources, approvals, versions, corrections, decisions, evidence, exceptions, and open gaps. Hand off to Product Marketing, Sales, Solutions, Documentation, Support, Legal, and Engineering.

RELATED CODEX SECTIONS

020-044 Architecture; 239-242 Performance and Scale Testing; 246 Security Testing; 287-288 Release Engineering and Management; 343 Product Marketing; 346-347 Sales and Solutions.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, architecture, testing, product, legal, support, and evidence context, and the integration roadmap. Exact claims, versions, benchmarks, reviewers, audiences, and acceptance authority remain unresolved.

345

SECTION 345 - BRAND MANAGEMENT

CODEX/03_SECTIONS/345_BRAND_MANAGEMENT.txt

SHA-256 a8b46bf1ebe0aedd02c353a4dd964d776b7f88e837caccca3c02c70f4b95bc92

4,772 BYTES / PRESERVED SOURCE IDENTITY

SECTION 345 — Brand Management

STATUS: CONTENT-COMPLETE

PURPOSE

Govern the identity, promise, expression, use, protection, accessibility, consistency, and evolution of a brand across products, services, channels, and relationships.

REQUIRED OUTCOME

A brand model with purpose, audience, promise, identity, assets, usage, rights, accessibility, governance, evidence, reputation, change, and recovery.

WHY THIS MATTERS

Brand communicates expectation and trust. Inconsistent, inaccessible, misleading, or culturally careless use can damage people and relationships while making product claims harder to interpret.

APPLICABILITY

Apply when a name, mark, visual or verbal identity, promise, public representation, or branded experience is created, changed, licensed, or retired.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no brand identity, promise, asset, audience, public representation, or trust expectation changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use product, user research, ethics, accessibility, IP, legal, marketing, localization, community, and communications baselines.

REQUIRED INPUTS

Purpose, audience, promise, identity, assets, usage, rights, channels, locale, accessibility, reputation, approvals, and evidence.

HUMAN RESPONSIBILITIES AND AUTHORITY

Brand owners approve promise, identity, usage, rights, cultural context, public response, exceptions, and retirement. Humans resolve reputation, harm, and ownership decisions.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft variants, analyze authorized feedback, check asset use, and identify inconsistency. It must not impersonate people, make public commitments, infer cultural acceptance, or publish brand material without human review.

APPLICABLE EXISTING AGENT ROLES

Brand Manager, Product Marketer, User Experience Reviewer, Researcher, IP Analyst, Legal Advisor, Accessibility Reviewer, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Define purpose and audience; create identity; verify rights and accessibility; govern use; localize; monitor reputation; respond; update; and retire or transfer assets.

HUMAN-AI INTERACTION REQUIREMENTS

Show asset, promise, audience, locale, source, rights, evidence, and limitation. Require human review for public, cultural, legal, accessibility, and crisis communication.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use asset registry, rights, version, accessibility, localization, approval, publication, takedown, and correction paths. Keep brand promise traceable to actual capability.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show asset, usage, audience, promise, evidence freshness, rights, incidents, owner, and authorized approve, correct, restrict, withdraw, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Brand charter; audience and promise; identity and asset register; rights and accessibility; usage and reputation; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify asset, promise, source, audience, locale, rights, method, time, result, reviewer, and limitation. Awareness or sentiment is bounded evidence, not reputation truth.

VERIFICATION AND ACCEPTANCE TESTS

Verify asset rights, consistency, accessibility, localization, product alignment, public use, cultural context, correction, and takedown. Test misleading or outdated claim.

FAILURE STATES AND ANTI-PATTERNS

Brand promise outruns product, inaccessible identity, unlicensed asset, cultural harm, inconsistent use, synthetic sentiment, and crisis response without accountable owner.

RECOVERY AND RECONSIDERATION

Withdraw or correct, communicate, protect affected people, preserve evidence, resolve rights, and reassess promise and use. Reopen on product, audience, culture, law, or reputation change.

COMPLETION CRITERIA

Purpose, audience, promise, assets, rights, accessibility, usage, evidence, owners, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve assets, approvals, rights, usage, incidents, communications, decisions, evidence, exceptions, and open gaps. Hand off to Product Marketing, Technical Marketing, IP, Legal, Product, Support, and Governance.

RELATED CODEX SECTIONS

219 Legal; 223 IP; 247 Accessibility; 250 Localization; 337 Community; 343-344 Product and Technical Marketing; 397-400 Retirement and Disposal.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, brand, product, user, IP, legal, accessibility, and reputation context, and the integration roadmap. Exact audience, promise, assets, rights, locales, approval, and acceptance authority remain unresolved.

346

SECTION 346 - SALES ENGINEERING

CODEX/03_SECTIONS/346_SALES_ENGINEERING.txt

SHA-256 31dc979c7c901673f00f5261930ecaa421887f90496bd0d2eb041d1cdf56b1de

5,024 BYTES / PRESERVED SOURCE IDENTITY

SECTION 346 — Sales Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Translate customer technical needs into truthful solution fit, architecture, demonstration, risk, integration, security, deployment, and implementation expectations.

REQUIRED OUTCOME

A sales-engineering model with customer context, requirements, proposed solution, assumptions, evidence, demo, integration, security, constraints, commitments, and handoff.

WHY THIS MATTERS

Sales discussions can create technical commitments before architecture and delivery review. A convincing demo may hide scale, security, data, support, or implementation limits.

APPLICABILITY

Apply when technical specialists support a customer or partner decision involving product fit, architecture, integration, security, deployment, or operational outcome.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no technical customer decision, solution claim, demo, integration, or commitment changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use product, solution, architecture, security, legal, IP, licensing, support, customer success, documentation, and roadmap baselines.

REQUIRED INPUTS

Customer need, context, constraints, architecture, data, security, integration, deployment, scale, evidence, demo, assumptions, and approval authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Sales engineers document facts and assumptions and escalate gaps. Product, architecture, security, legal, and delivery owners approve commitments, exceptions, and customer-specific claims.

AI RESPONSIBILITIES AND BOUNDARIES

AI may summarize authorized customer context, draft diagrams and questions, and compare approved capabilities. It must not invent fit, guarantee performance, expose customer data, make commitments, or impersonate technical authority.

APPLICABLE EXISTING AGENT ROLES

Sales Engineer, Solutions Engineer, Architect, Researcher, Analyst, Security Auditor, Product Manager, Legal Advisor, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Understand need; qualify constraints; map requirements; design and review solution; demonstrate; validate integration, security, data, scale, operations, and support; state assumptions; obtain approvals; and hand off.

HUMAN-AI INTERACTION REQUIREMENTS

Show customer, need, capability, version, evidence, assumption, limitation, reviewer, and commitment status. Require human approval for customer claim, architecture, pricing, legal, security, and delivery promise.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link customer requirement to product capability, architecture, data, security, integration, deployment, support, and evidence. Separate demo state from production state.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show opportunity, customer need, solution, assumptions, evidence freshness, risks, owners, commitment status, and authorized approve, qualify, escalate, correct, and handoff controls.

REQUIRED OUTPUTS AND ARTIFACTS

Technical discovery; requirement and constraint map; solution architecture; demo and evidence; assumption and risk; security, deployment, support, decision, and handoff records.

EVIDENCE REQUIREMENTS

Identify customer, need, capability, version, environment, demo, source, method, result, assumption, and limitation. Demo success proves neither production fit nor customer outcome.

VERIFICATION AND ACCEPTANCE TESTS

Verify requirement fit, architecture, integration, security, data, performance, deployment, support, licensing, assumptions, and handoff. Test customer scenario and failure path.

FAILURE STATES AND ANTI-PATTERNS

Demo as proof, unapproved promise, customer data in demo, architecture hand-waving, hidden assumption, security claim without evidence, and technical debt transferred silently.

RECOVERY AND RECONSIDERATION

Correct customer record, qualify or withdraw claim, notify stakeholders, preserve evidence, and reassess solution and commitment. Reopen on product, customer, architecture, security, or delivery change.

COMPLETION CRITERIA

Customer, need, solution, evidence, assumptions, risks, approvals, limitations, commitment, owners, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve discovery, diagrams, demos, commitments, assumptions, decisions, evidence, exceptions, and open gaps. Hand off to Solutions, Product, Architecture, Engineering, Delivery, Support, Legal, and Customer Success.

RELATED CODEX SECTIONS

019 Requirements; 022 Solution Architecture; 010 Product Management; 219 Legal; 224 Licensing; 343-344 Marketing; 347 Solutions Engineering; 336 Customer Success.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, customer, product, solution, architecture, legal, security, and delivery context, and the integration roadmap. Exact customers, capabilities, evidence, commitments, security, support, and acceptance authority remain unresolved.

347

SECTION 347 - SOLUTIONS ENGINEERING

CODEX/03_SECTIONS/347_SOLUTIONS_ENGINEERING.txt

SHA-256 d96cd5d2aa5fd799ff733a7526c15dbb91827d4ad3abcaf166737792bfa95ad2

5,331 BYTES / PRESERVED SOURCE IDENTITY

SECTION 347 — Solutions Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Design and validate implementable customer solutions that connect business outcomes, product capability, architecture, integration, security, operations, delivery, and ownership.

REQUIRED OUTCOME

A solution model with customer outcome, requirements, architecture, data, integration, security, deployment, operations, cost, risks, assumptions, evidence, and handoff.

WHY THIS MATTERS

A solution can satisfy a diagram while failing implementation, support, security, data, cost, or customer adoption. The handoff must preserve assumptions and unresolved work rather than turning proposal into fact.

APPLICABILITY

Apply when a customer, partner, or internal stakeholder needs a composed technical and operational solution across products, teams, providers, or domains.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no composed solution, architecture, integration, delivery, customer outcome, or ownership claim changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use requirements, solution architecture, product, domain, data, security, privacy, integration, operations, cost, legal, licensing, delivery, and support baselines.

REQUIRED INPUTS

Outcome, stakeholders, requirements, constraints, current state, products, architecture, data, integrations, security, deployment, operations, cost, risks, and acceptance.

HUMAN RESPONSIBILITIES AND AUTHORITY

Solution owners coordinate specialists and preserve customer truth. Accountable product, architecture, security, legal, delivery, operations, and customer owners approve commitments and tradeoffs.

AI RESPONSIBILITIES AND BOUNDARIES

AI may synthesize authorized requirements and sources, draft options and diagrams, identify gaps, and compare bounded designs. It must not claim feasibility, make commitments, expose customer data, or replace specialist or owner approval.

APPLICABLE EXISTING AGENT ROLES

Solutions Engineer, Architect, Business Analyst, Researcher, Analyst, Security Auditor, Data Engineer, Operations Engineer, Verifier, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Discover outcome and constraints; inspect current state; model requirements and dependencies; design options; assess security, data, cost, delivery, operations, support, and lifecycle; validate; decide; and hand off.

HUMAN-AI INTERACTION REQUIREMENTS

Show outcome, stakeholder, option, evidence, assumption, risk, owner, commitment, and limitation. Require human approval for architecture, security, data, legal, pricing, deployment, and customer claims.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link outcome to capabilities, components, data, interfaces, identity, security, deployment, operation, support, recovery, cost, and retirement. Preserve alternative and rejected option rationale.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show solution state, requirements, options, risks, assumptions, evidence freshness, owners, decisions, and authorized approve, revise, qualify, escalate, reject, and handoff controls.

REQUIRED OUTPUTS AND ARTIFACTS

Solution charter; outcome and requirement map; architecture and integration; data and security; deployment and operations; cost, risk, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify outcome, requirement, source, product, version, environment, method, result, assumption, owner, and limitation. A diagram or demo proves neither feasibility nor production outcome.

VERIFICATION AND ACCEPTANCE TESTS

Verify requirements, architecture, integration, data, identity, security, privacy, cost, delivery, operations, support, recovery, and customer understanding. Reproduce material assumptions.

FAILURE STATES AND ANTI-PATTERNS

Solution theater, product fit assumed, customer data exposed, unsupported architecture, handoff without owner, hidden operations, and proposal presented as committed implementation.

RECOVERY AND RECONSIDERATION

Mark uncertainty, correct customer and internal records, preserve decisions, revise or reject option, and reassess commitments and downstream work. Reopen on requirement, product, architecture, customer, risk, or provider change.

COMPLETION CRITERIA

Outcome, requirements, architecture, options, dependencies, evidence, risks, owners, commitments, operations, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending stakeholder acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve discovery, requirements, diagrams, assumptions, decisions, evidence, commitments, exceptions, and open gaps. Hand off to Product, Architecture, Engineering, Delivery, Operations, Support, Security, Legal, and Customer Success.

RELATED CODEX SECTIONS

019 Requirements; 022 Solution Architecture; 021 Enterprise Architecture; 010 Product Management; 346 Sales Engineering; 273 Developer Experience; 301-331 Operations and Readiness.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, customer, requirements, architecture, security, delivery, operations, legal, and support context, and the integration roadmap. Exact stakeholders, outcome, products, costs, risks, commitments, implementation evidence, and acceptance authority remain unresolved.

348

SECTION 348 - PROCUREMENT

CODEX/03_SECTIONS/348_PROCUREMENT.txt

SHA-256 2a8664019bd4e09a43b63d11f339b26a77006620a2211412b34d1583e81561ba

5,020 BYTES / PRESERVED SOURCE IDENTITY

SECTION 348 — Procurement

STATUS: CONTENT-COMPLETE

PURPOSE

Acquire goods, services, software, hardware, data, facilities, and suppliers through a fair, authorized, evidence-based process that protects value, rights, security, continuity, and people.

REQUIRED OUTCOME

A procurement model with need, specification, market, evaluation, authority, conflict, price, contract, security, privacy, IP, delivery, acceptance, supplier risk, and exit.

WHY THIS MATTERS

Procurement choices create long-lived technical, financial, legal, and operational dependencies. Lowest price, polished proposal, or urgent demand does not prove suitability, integrity, resilience, or total cost.

APPLICABILITY

Apply when an organization selects, buys, renews, leases, contracts, or materially changes a supplier, product, service, asset, or license.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no purchase, supplier, contract, asset, cost, obligation, or dependency changes and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use strategy, requirements, legal, IP, licensing, security, privacy, architecture, finance, supply-chain, quality, and vendor baselines.

REQUIRED INPUTS

Need, specification, budget, alternatives, suppliers, evaluation, conflicts, contract, security, privacy, delivery, acceptance, support, continuity, and exit.

HUMAN RESPONSIBILITIES AND AUTHORITY

Authorized procurement, finance, legal, security, technical, privacy, and business owners evaluate and approve. Humans manage conflicts, negotiate, accept risk, sign, and terminate.

AI RESPONSIBILITIES AND BOUNDARIES

AI may compare authorized proposals, extract terms, draft questions, and identify gaps. It must not select supplier, negotiate, make representations, expose confidential bids, or approve spend or contract without human authority.

APPLICABLE EXISTING AGENT ROLES

Procurement Analyst, Business Analyst, Architect, Legal Advisor, Security Auditor, Privacy Reviewer, Financial Analyst, Researcher, Verifier, and Governance Reviewer.

REQUIRED ACTIVITIES

Define need; specify; research; solicit; evaluate; check conflicts and rights; assess supplier risk; negotiate; approve; contract; receive; accept; monitor; renew or exit.

HUMAN-AI INTERACTION REQUIREMENTS

Show supplier, requirement, proposal, source, evaluation, conflict, cost, risk, evidence, and limitation. Require human approval for selection, spend, negotiation, contract, exception, and termination.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Capture supplier, product, dependency, data, security, support, SLA, license, provenance, integration, portability, and exit as linked records.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show procurement status, suppliers, evaluation, spend, contract, risks, evidence freshness, owner, and authorized approve, compare, award, hold, renew, remediate, and terminate controls.

REQUIRED OUTPUTS AND ARTIFACTS

Procurement charter; requirement and evaluation; supplier and conflict; contract and risk; delivery and acceptance; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify need, supplier, proposal, version, price, source, evaluator, method, time, result, conflict, and limitation. Proposal or certification proves only the supplier's bounded claim.

VERIFICATION AND ACCEPTANCE TESTS

Verify requirements, supplier identity, conflicts, price, contract, security, privacy, IP, delivery, quality, support, continuity, acceptance, and exit. Test nonperformance and substitution.

FAILURE STATES AND ANTI-PATTERNS

Unapproved spend, sole-source without reason, conflict hidden, price-only decision, contract not linked to architecture, supplier claim as evidence, and exit impossible.

RECOVERY AND RECONSIDERATION

Pause purchase or use, preserve records, escalate conflict or nonperformance, apply remedy or substitute, protect data, and reassess dependency and cost. Reopen on supplier, product, contract, risk, or requirement change.

COMPLETION CRITERIA

Need, specification, authority, suppliers, evaluation, cost, contract, risks, acceptance, evidence, owners, exit, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending authorized approval.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve procurement, proposals, evaluations, contracts, suppliers, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Vendor, Supply Chain, Legal, Finance, Security, Architecture, and Operations.

RELATED CODEX SECTIONS

219 Legal; 223-224 IP and Licensing; 260 Dependency; 349-350 Vendor and Supply Chain; 354 Hardware Certification; 358-364 Commerce and Finance; 397-400 Retirement.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, procurement, legal, financial, supplier, security, and architecture context, and the integration roadmap. Exact authority, thresholds, markets, suppliers, terms, conflicts, and acceptance evidence remain unresolved.

349

SECTION 349 - VENDOR MANAGEMENT

CODEX/03_SECTIONS/349_VENDOR_MANAGEMENT.txt

SHA-256 411d61b79a301db683ac04d0245375cd9ab8d3262ccd59dcf3943efbd31bbd24

4,805 BYTES / PRESERVED SOURCE IDENTITY

SECTION 349 — Vendor Management

STATUS: CONTENT-COMPLETE

PURPOSE

Govern supplier relationships, performance, risk, security, privacy, contracts, service levels, changes, incidents, continuity, and exit throughout the vendor lifecycle.

REQUIRED OUTCOME

A vendor model with identity, service, contract, owner, performance, risk, data, access, evidence, review, issue, remediation, renewal, and exit.

WHY THIS MATTERS

Vendor dependency persists beyond purchase. A supplier may change ownership, subprocessor, control, price, support, region, or viability without a local code change.

APPLICABILITY

Apply when an external supplier provides goods, service, data, software, infrastructure, labor, support, logistics, or operational dependency.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no vendor, contract, service, data, dependency, risk, or relationship behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use procurement, legal, privacy, security, architecture, supply chain, service-level, finance, continuity, and exit baselines.

REQUIRED INPUTS

Vendor, service, contract, owner, contacts, data, access, SLA, controls, performance, incidents, financial health, changes, and exit.

HUMAN RESPONSIBILITIES AND AUTHORITY

Vendor owners monitor and communicate; legal, security, privacy, finance, and service owners review their concerns. Humans approve renewal, exception, remediation, notice, and termination.

AI RESPONSIBILITIES AND BOUNDARIES

AI may organize authorized vendor evidence, compare performance, draft questions, and identify change signals. It must not certify supplier, negotiate, disclose confidential information, accept risk, or terminate without authority.

APPLICABLE EXISTING AGENT ROLES

Vendor Manager, Procurement Analyst, Legal Advisor, Security Auditor, Privacy Reviewer, Service Manager, Financial Analyst, Researcher, Verifier, and Governance Reviewer.

REQUIRED ACTIVITIES

Inventory vendors; review contract and controls; monitor performance, incidents, data, and change; assess risk and financial health; remediate; renew; substitute; and exit.

HUMAN-AI INTERACTION REQUIREMENTS

Show vendor, service, contract, evidence, performance, risk, data, change, owner, and limitation. Require human approval for renewal, exception, disclosure, escalation, and termination.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link vendor to services, data, access, dependencies, SLAs, subvendors, regions, artifacts, continuity, and exit. Minimize lock-in and retain alternate path where consequence warrants.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show vendor health, service, contract, performance, incidents, evidence freshness, data, owner, and authorized remediate, restrict, renew, substitute, and terminate controls.

REQUIRED OUTPUTS AND ARTIFACTS

Vendor register; service and contract; risk and control; performance and incident; renewal and exit; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify vendor, service, contract, period, source, method, result, owner, data, incident, and limitation. Vendor attestation or dashboard proves only its stated scope.

VERIFICATION AND ACCEPTANCE TESTS

Verify identity, contract, service, controls, data, access, SLA, incident, continuity, subvendor, change notice, financial viability, and exit. Test vendor outage and nonperformance.

FAILURE STATES AND ANTI-PATTERNS

No owner, contract not operationalized, vendor claim as proof, data or subprocessor hidden, renewal by default, lock-in, and incident communication delayed.

RECOVERY AND RECONSIDERATION

Restrict access, preserve evidence, invoke remedy, substitute or fail over, notify stakeholders, protect data, and reassess relationship. Reopen on vendor, service, contract, control, or viability change.

COMPLETION CRITERIA

Vendor, service, contract, owner, data, access, performance, risk, evidence, remediation, renewal, exit, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve vendor records, contracts, assessments, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Procurement, Supply Chain, Legal, Security, Privacy, Service Management, and Operations.

RELATED CODEX SECTIONS

348 Procurement; 350 Supply Chain; 219 Legal; 220 Compliance; 217 Trust and Safety; 326-330 Continuity and Recovery; 397-400 Decommissioning.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, vendor, legal, security, privacy, service, and continuity context, and the integration roadmap. Exact vendors, contracts, data, subvendors, metrics, remedies, exit, and acceptance authority remain unresolved.

350

SECTION 350 - SUPPLY CHAIN MANAGEMENT

CODEX/03_SECTIONS/350_SUPPLY_CHAIN_MANAGEMENT.txt

SHA-256 f20e55aea6f2065a42096856e7202ed095ef88b5b99ce2b25088d6a8ff49ff3a

4,999 BYTES / PRESERVED SOURCE IDENTITY

SECTION 350 — Supply Chain Management

STATUS: CONTENT-COMPLETE

PURPOSE

Plan, source, qualify, monitor, protect, deliver, and recover the flow of materials, components, software, data, services, and suppliers across the lifecycle.

REQUIRED OUTCOME

A supply-chain model with tiers, provenance, demand, inventory, criticality, suppliers, controls, logistics, quality, disruption, substitution, evidence, and retirement.

WHY THIS MATTERS

Supply chains are distributed trust systems. Counterfeit, compromised, delayed, substituted, single-sourced, or untraceable inputs can affect safety, security, quality, cost, and continuity.

APPLICABILITY

Apply when a product, service, system, or operation depends on multi-tier materials, components, software, data, logistics, suppliers, or external capacity.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no supply input, tier, supplier, inventory, delivery, quality, or continuity behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use procurement, vendor, hardware, software supply-chain, quality, security, logistics, manufacturing, finance, continuity, and risk baselines.

REQUIRED INPUTS

Demand, materials, components, software, suppliers, tiers, provenance, inventory, lead time, quality, risk, logistics, alternatives, and disruption plans.

HUMAN RESPONSIBILITIES AND AUTHORITY

Supply-chain owners approve sources, substitutions, criticality, inventory, ethical and safety requirements, disruption response, cost, and risk. Humans resolve supplier conflicts and acceptance.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized supply data, identify concentration or delay risk, and draft alternatives. It must not select or substitute critical input, expose supplier data, make safety claims, or alter orders without authority.

APPLICABLE EXISTING AGENT ROLES

Supply Chain Manager, Procurement Analyst, Vendor Manager, Hardware Engineer, Security Auditor, Quality Engineer, Researcher, Analyst, Verifier, and Planner.

REQUIRED ACTIVITIES

Map tiers and critical inputs; qualify sources; capture provenance; plan demand and inventory; monitor quality and risk; manage logistics; prepare alternatives; respond to disruption; and retire sources.

HUMAN-AI INTERACTION REQUIREMENTS

Show input, tier, supplier, provenance, criticality, inventory, evidence, risk, alternative, and limitation. Require human approval for substitution, safety, security, cost, and disruption action.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use traceability, approved source, component and package identity, dual source where warranted, inventory, quality gate, secure logistics, substitution, and continuity.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show supply graph, critical items, source, inventory, lead time, quality, risk, evidence freshness, owner, and authorized qualify, substitute, expedite, quarantine, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Supply-chain charter; tier and provenance map; demand and inventory; supplier and quality; disruption and alternative; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify input, tier, source, supplier, lot or version, method, time, result, owner, and limitation. Supplier declaration or inventory count proves only its bounded claim.

VERIFICATION AND ACCEPTANCE TESTS

Verify provenance, source, quality, inventory, lead time, substitution, security, safety, logistics, supplier, continuity, and acceptance. Test counterfeit, delay, shortage, and supplier loss.

FAILURE STATES AND ANTI-PATTERNS

Single source, hidden tier, untraceable component, substitution without review, inventory as quality, supplier concentration ignored, and cost-only choice.

RECOVERY AND RECONSIDERATION

Quarantine affected input, preserve provenance, use approved alternative, communicate, reassess safety and quality, and update design or inventory. Reopen on source, tier, supplier, demand, or threat change.

COMPLETION CRITERIA

Inputs, tiers, suppliers, provenance, inventory, quality, risks, alternatives, evidence, owners, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve supply graph, sources, substitutions, quality, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Procurement, Vendor, Manufacturing, Logistics, Security, Quality, and Operations.

RELATED CODEX SECTIONS

199 Hardware Security; 209 Vulnerability; 223-224 IP and Licensing; 348 Procurement; 349 Vendor; 351-357 Manufacturing, Logistics, and Distribution; 326 Continuity.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, supply, procurement, hardware, software, quality, and continuity context, and the integration roadmap. Exact tiers, sources, criticality, inventory, ethics, alternatives, and acceptance authority remain unresolved.

351

SECTION 351 - MANUFACTURING

CODEX/03_SECTIONS/351_MANUFACTURING.txt

SHA-256 968f3333eff4d4cf58881c2f14546445e25d8d9ad0b0965a7568f7600cd1a1b8

5,034 BYTES / PRESERVED SOURCE IDENTITY

SECTION 351 — Manufacturing

STATUS: CONTENT-COMPLETE

PURPOSE

Transform approved materials, components, instructions, software, labor, and equipment into consistent, traceable, safe, and accepted products or outputs.

REQUIRED OUTCOME

A manufacturing model with process, inputs, equipment, people, quality, traceability, configuration, work instructions, safety, capacity, inspection, nonconformance, and release.

WHY THIS MATTERS

Manufacturing variation can turn a correct design into inconsistent, unsafe, counterfeit, or untraceable output. Schedule and yield cannot replace quality, safety, provenance, and worker protection.

APPLICABILITY

Apply when physical or repeatable production transforms materials, components, firmware, software, data, or assemblies into deliverables.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no production process, input, quality, traceability, safety, or output behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use product, hardware, embedded, supply chain, quality, safety, procurement, facility, workforce, testing, and logistics baselines.

REQUIRED INPUTS

Design, bill of materials, work instructions, inputs, equipment, process, operators, quality, safety, capacity, traceability, inspection, and acceptance.

HUMAN RESPONSIBILITIES AND AUTHORITY

Manufacturing and quality owners approve process, materials, substitutions, safety, release, nonconformance, and capacity. Workers can stop unsafe work and report defects.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized production data, identify variation, draft instructions and inspections, and propose scheduling. It must not change process, approve lot, infer worker performance, or control equipment without qualified human authority.

APPLICABLE EXISTING AGENT ROLES

Manufacturing Engineer, Quality Engineer, Hardware Engineer, Safety Reviewer, Supply Chain Manager, Planner, Tester, Verifier, Observer, and Documentation Writer.

REQUIRED ACTIVITIES

Define process and inputs; qualify equipment and operators; issue instructions; produce; inspect; trace; handle nonconformance; protect safety; measure yield and quality; release; and improve or retire process.

HUMAN-AI INTERACTION REQUIREMENTS

Show product, lot, process, input, equipment, operator role, result, safety, evidence, and limitation. Require human review for substitution, quality release, safety, and worker-impact decisions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use revision-controlled instructions, lot traceability, equipment calibration, segregated nonconformance, quality gates, safety interlocks, secure provisioning, and recall path.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show process, lot, input, equipment, quality, safety, capacity, evidence freshness, owner, and authorized hold, inspect, quarantine, release, rework, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Manufacturing charter; process and work instruction; input and lot; quality and safety; nonconformance and release; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify product, lot, input, process, equipment, revision, operator role, method, time, result, inspector, and limitation. Yield or inspection sample proves only its declared population and method.

VERIFICATION AND ACCEPTANCE TESTS

Verify process, input, equipment, calibration, traceability, quality, safety, software or firmware, inspection, nonconformance, release, and recall. Test variation and equipment failure.

FAILURE STATES AND ANTI-PATTERNS

Unapproved substitution, traceability gap, unsafe work, calibration expired, sample theater, quality release by schedule, and defect or rework hidden.

RECOVERY AND RECONSIDERATION

Stop or quarantine, preserve lot and process evidence, inspect or rework, notify owners, recall where needed, and reassess safety and downstream impact. Reopen on process, input, equipment, or design change.

COMPLETION CRITERIA

Process, inputs, equipment, people, safety, quality, traceability, release, evidence, owners, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve work instructions, revisions, lots, equipment, inspections, nonconformance, decisions, evidence, exceptions, and open gaps. Hand off to Production Planning, Assembly, Certification, Provisioning, Logistics, Quality, and Operations.

RELATED CODEX SECTIONS

058-060 Hardware, Embedded, and Firmware; 350 Supply Chain; 352-357 Production, Assembly, Certification, Provisioning, Logistics, and Distribution; 232 Hardware Testing; 252 Safety Testing.

CODEX/03_SECTIONS/351_MANUFACTURING.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, hardware, supply, quality, safety, workforce, and logistics context, and the integration roadmap. Exact products, facilities, standards, process, workforce, lot, safety, and acceptance authority remain unresolved.

352

SECTION 352 - PRODUCTION PLANNING

CODEX/03_SECTIONS/352_PRODUCTION_PLANNING.txt

SHA-256 9498c83fe0ac6c84a0b50cbbfb624d2bbbd53cad8f163d7134386cf69b9119fb

4,777 BYTES / PRESERVED SOURCE IDENTITY

SECTION 352 — Production Planning

STATUS: CONTENT-COMPLETE

PURPOSE

Translate demand, design, capacity, materials, labor, equipment, quality, safety, and delivery commitments into an executable and adaptable production plan.

REQUIRED OUTCOME

A production plan with demand, priority, materials, capacity, labor, equipment, sequence, quality, safety, constraints, contingency, evidence, and owner decision.

WHY THIS MATTERS

Plans trade among delivery, cost, quality, safety, workers, inventory, and resilience. A schedule can be feasible on paper while relying on unavailable inputs or unsafe load.

APPLICABILITY

Apply when production, assembly, manufacturing, provisioning, service capacity, or delivery is planned or materially re-sequenced.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no demand, production, capacity, material, workforce, safety, or delivery behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use product, manufacturing, supply chain, capacity, workforce, safety, quality, logistics, finance, and customer commitment baselines.

REQUIRED INPUTS

Demand, product, materials, inventory, capacity, labor, equipment, sequence, quality, safety, lead time, commitments, constraints, and alternatives.

HUMAN RESPONSIBILITIES AND AUTHORITY

Planning owners approve priority, schedule, overtime, substitution, safety, customer commitment, cost, and contingency. Humans resolve competing demand and worker impact.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized demand and capacity, draft schedules, and identify constraints. It must not commit delivery, change priority, infer worker availability, or substitute material without authority.

APPLICABLE EXISTING AGENT ROLES

Production Planner, Supply Chain Manager, Manufacturing Engineer, Capacity Analyst, Quality Engineer, Safety Reviewer, Researcher, Analyst, and Verifier.

REQUIRED ACTIVITIES

Forecast demand; map materials and capacity; set priority; schedule; validate constraints; coordinate production; monitor; adjust; manage exception; communicate; and review plan accuracy.

HUMAN-AI INTERACTION REQUIREMENTS

Show demand, product, material, capacity, labor, equipment, priority, commitment, evidence, and limitation. Require human approval for schedule, safety, overtime, substitution, and customer promise.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use versioned plan, inventory and capacity data, constraint, quality gate, safety, traceability, change, alternative, and contingency modeling.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show demand, schedule, inventory, capacity, constraints, quality, safety, evidence freshness, owner, and authorized prioritize, reschedule, hold, substitute, expedite, and cancel controls.

REQUIRED OUTPUTS AND ARTIFACTS

Production plan; demand and capacity; materials and sequence; quality and safety; exception and contingency; decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify demand, product, plan version, material, capacity, source, method, time, result, owner, and limitation. Forecast or schedule proves intent, not execution or delivery.

VERIFICATION AND ACCEPTANCE TESTS

Verify constraints, materials, capacity, labor, equipment, safety, quality, schedule, contingency, customer commitment, and plan changes. Test shortage and equipment failure.

FAILURE STATES AND ANTI-PATTERNS

Schedule without material, safety or quality sacrificed, forecast as fact, hidden overtime, single-source plan, priority drift, and customer promise without evidence.

RECOVERY AND RECONSIDERATION

Replan, protect safety and quality, communicate, use approved alternative, preserve decisions, and reassess commitments. Reopen on demand, input, capacity, workforce, equipment, or product change.

COMPLETION CRITERIA

Demand, plan, materials, capacity, people, equipment, quality, safety, evidence, owners, contingencies, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve plans, versions, demand, exceptions, decisions, evidence, safety, quality, and open gaps. Hand off to Manufacturing, Assembly, Supply Chain, Logistics, Finance, Sales, and Operations.

RELATED CODEX SECTIONS

018 Risk; 228 Quality; 321 Capacity; 350-353 Supply Chain, Manufacturing, Planning, and Assembly; 356-357 Logistics and Distribution; 358 Commerce.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, production, supply, capacity, quality, safety, and commerce context, and the integration roadmap. Exact demand, products, inventory, workforce, equipment, priorities, commitments, and acceptance authority remain unresolved.

353

SECTION 353 - ASSEMBLY

CODEX/03_SECTIONS/353_ASSEMBLY.txt

SHA-256 bef7b445a793df0c4c698a65e324aa20adc8c9e527a0a04edcece2aa6820e7e9

4,656 BYTES / PRESERVED SOURCE IDENTITY

SECTION 353 — Assembly

STATUS: CONTENT-COMPLETE

PURPOSE

Combine approved components, software, firmware, materials, and instructions into a traceable, safe, tested, and accepted assembly.

REQUIRED OUTCOME

An assembly model with component identity, revision, sequence, tools, operators, interfaces, configuration, inspection, quality, safety, test, rework, and release.

WHY THIS MATTERS

Assembly errors can be invisible until field use. A component can be correct in isolation while orientation, connection, version, torque, firmware, or configuration makes the whole unsafe.

APPLICABILITY

Apply when parts, modules, software, firmware, or services are combined into a product, device, system, kit, or deployable unit.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no assembly, integration, component traceability, quality, safety, or output behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use manufacturing, hardware, embedded, supply, quality, safety, testing, provisioning, and logistics baselines.

REQUIRED INPUTS

Assembly definition, components, revisions, tools, sequence, operator, configuration, software, firmware, inspection, test, safety, and acceptance.

HUMAN RESPONSIBILITIES AND AUTHORITY

Assembly and quality owners approve parts, sequence, substitution, inspection, rework, safety, and release. Workers can stop unsafe or suspect assembly.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized assembly records, identify mismatch, draft checklists, and analyze quality. It must not control equipment, approve lot, infer worker fault, or change sequence without qualified human authority.

APPLICABLE EXISTING AGENT ROLES

Assembly Engineer, Manufacturing Engineer, Hardware Engineer, Embedded Engineer, Quality Engineer, Safety Reviewer, Tester, Verifier, and Observer.

REQUIRED ACTIVITIES

Verify parts and revision; prepare tools; assemble; configure; inspect; test; trace; quarantine; rework; release; package; and capture lessons.

HUMAN-AI INTERACTION REQUIREMENTS

Show unit, part, revision, sequence, tool, operator role, configuration, result, safety, and limitation. Require human approval for substitution, rework, release, and safety.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use work instructions, serialization, component traceability, poka-yoke or equivalent controls, configuration binding, calibration, inspection, test, and rework segregation.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show unit, parts, revisions, progress, quality, safety, evidence freshness, owner, and authorized hold, inspect, rework, quarantine, release, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Assembly charter; unit and component record; work instruction; inspection and test; rework and release; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify unit, part, revision, sequence, tool, configuration, method, time, result, inspector, and limitation. Assembly completion proves neither correct function nor safety.

VERIFICATION AND ACCEPTANCE TESTS

Verify component, orientation, connection, software, firmware, configuration, inspection, quality, safety, function, rework, traceability, and release. Test variation and misassembly.

FAILURE STATES AND ANTI-PATTERNS

Wrong revision, untraceable unit, skipped inspection, unsafe rework, tool calibration ignored, assembly count as quality, and component pass treated as system pass.

RECOVERY AND RECONSIDERATION

Quarantine unit and related lot, preserve evidence, inspect or rework, notify owners, retest, and reassess downstream units. Reopen on component, instruction, tool, design, or quality change.

COMPLETION CRITERIA

Unit, parts, revisions, sequence, tools, quality, safety, tests, evidence, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve assembly records, units, parts, tests, rework, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Manufacturing, Certification, Provisioning, Logistics, Quality, and Operations.

RELATED CODEX SECTIONS

058-060 Hardware, Embedded, and Firmware; 351-352 Manufacturing and Planning; 354 Certification; 355 Provisioning; 232 Hardware Testing; 252 Safety Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, hardware, manufacturing, quality, safety, testing, and logistics context, and the integration roadmap. Exact units, components, tools, standards, workers, safety, and acceptance authority remain unresolved.

354

SECTION 354 - HARDWARE CERTIFICATION

CODEX/03_SECTIONS/354_HARDWARE_CERTIFICATION.txt

SHA-256 558b5f5d8b65202fae01209e82ede366318ee24a8118dfa67638a815a1f556f7

5,163 BYTES / PRESERVED SOURCE IDENTITY

SECTION 354 — Hardware Certification

STATUS: CONTENT-COMPLETE

PURPOSE

Establish whether a hardware product or component meets a declared certification, conformity, safety, electromagnetic, environmental, or sector requirement for its exact scope.

REQUIRED OUTCOME

A certification model with authority, standard, product identity, configuration, evidence, test laboratory, samples, findings, conditions, validity, marking, surveillance, and change impact.

WHY THIS MATTERS

Certification is bounded by standard, version, sample, configuration, jurisdiction, and validity. A certificate can be misapplied to a changed product or treated as total safety or security.

APPLICABILITY

Apply when a hardware product, device, component, facility, or process requires formal certification, conformity, approval, registration, or regulated evidence.

NON-APPLICABILITY CONDITIONS

Record non-applicability only after qualified review of product, market, jurisdiction, use, standard, and change scope confirms no certification or conformity claim is affected.

PREREQUISITES AND DEPENDENCIES

Use legal, regulatory, safety, hardware, manufacturing, testing, quality, supply, documentation, and change baselines.

REQUIRED INPUTS

Product, model, revision, standard, jurisdiction, intended use, samples, configuration, tests, laboratory, evidence, findings, validity, and change process.

HUMAN RESPONSIBILITIES AND AUTHORITY

Qualified certification, legal, safety, quality, and engineering owners approve scope, laboratory, submission, findings, marking, change, surveillance, and release. Humans attest and communicate boundaries.

AI RESPONSIBILITIES AND BOUNDARIES

AI may organize authorized certification evidence, compare requirements, draft questions, and identify changed configuration. It must not certify, sign, select laboratory, conceal finding, or claim regulatory approval without qualified human authority.

APPLICABLE EXISTING AGENT ROLES

Certification Engineer, Hardware Engineer, Safety Reviewer, Legal Advisor, Compliance Tester, Quality Assurance Reviewer, Researcher, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Identify authority and standard; define product and configuration; select lab; prepare samples; test; address findings; submit; receive; mark and communicate; monitor validity; manage change; and renew or retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show product, revision, standard, jurisdiction, sample, lab, evidence, validity, finding, and limitation. Require qualified human review for submission, attestation, marking, public claim, and change impact.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Bind certification to model, configuration, components, firmware, manufacturing process, and intended use. Keep certificate, evidence, change, and release records linked.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show certification, scope, product, revision, jurisdiction, validity, findings, evidence freshness, owner, and authorized submit, hold, change, mark, renew, withdraw, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Certification charter; authority and standard; product and configuration; test and lab; findings and submission; certificate and change impact; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify authority, standard, version, product, configuration, sample, lab, method, time, result, validity, and limitation. Certificate proves only its declared scope and conditions.

VERIFICATION AND ACCEPTANCE TESTS

Verify identity, configuration, standard, samples, laboratory, tests, findings, validity, marking, change notification, surveillance, and market scope. Test changed component or firmware impact.

FAILURE STATES AND ANTI-PATTERNS

Certificate applied to different revision, expired approval, unqualified lab, finding hidden, certification as total security, change without review, and marking beyond scope.

RECOVERY AND RECONSIDERATION

Hold or withdraw affected product, preserve evidence, notify authority and customers as required, assess field impact, remediate or recertify, and correct claims. Reopen on product, standard, jurisdiction, or change.

COMPLETION CRITERIA

Authority, standard, product, configuration, evidence, lab, validity, findings, owners, change impact, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve certificates, submissions, tests, labs, configurations, findings, decisions, evidence, exceptions, and open gaps. Hand off to Manufacturing, Assembly, Safety, Legal, Compliance, Release, and Support.

RELATED CODEX SECTIONS

050 Safety Architecture; 199 Hardware Security; 232 Hardware Testing; 251 Compliance Testing; 252 Safety Testing; 351 Manufacturing; 353 Assembly; 219 Legal.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, hardware, safety, legal, regulatory, quality, and testing context, and the integration roadmap. Exact products, standards, jurisdictions, laboratories, validity, changes, and acceptance authority remain unresolved.

355

SECTION 355 - DEVICE PROVISIONING

CODEX/03_SECTIONS/355_DEVICE_PROVISIONING.txt

SHA-256 20557950f0954d6cf2a93f528625d141bc4a93743b35dc6213b1b3becb989b15

4,953 BYTES / PRESERVED SOURCE IDENTITY

SECTION 355 — Device Provisioning

STATUS: CONTENT-COMPLETE

PURPOSE

Initialize devices with identity, firmware, configuration, credentials, ownership, policy, data, connectivity, and support so they enter service in a known and recoverable state.

REQUIRED OUTCOME

A device-provisioning model with asset identity, hardware and firmware, keys, owner, configuration, enrollment, security, data, validation, handoff, update, wipe, and retirement.

WHY THIS MATTERS

Provisioning establishes device trust and ownership. Shared credentials, wrong configuration, incomplete enrollment, exposed keys, or untracked handoff can compromise an entire fleet.

APPLICABILITY

Apply when a device, embedded system, endpoint, gateway, or hardware unit is prepared, enrolled, assigned, re-provisioned, transferred, or recovered.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no device identity, configuration, ownership, credential, firmware, enrollment, or lifecycle behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use hardware, embedded security, IAM, keys, secrets, configuration, manufacturing, inventory, update, privacy, support, and disposal baselines.

REQUIRED INPUTS

Device, hardware, firmware, identity, keys, owner, configuration, policy, network, data, enrollment, validation, support, wipe, and retirement.

HUMAN RESPONSIBILITIES AND AUTHORITY

Device owners approve identity, assignment, policy, data, key, enrollment, exception, transfer, wipe, recovery, and retirement. Humans verify chain of custody and safety.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized provisioning metadata and draft checklists or validation. It must not generate or expose keys, assign ownership, enroll, configure, wipe, or declare trust without authority.

APPLICABLE EXISTING AGENT ROLES

Device Provisioning Engineer, Embedded Engineer, IAM Engineer, Key Manager, Security Auditor, Manufacturing Engineer, Tester, Verifier, and Operations Engineer.

REQUIRED ACTIVITIES

Verify asset; provision identity and keys; install firmware; configure; enroll; validate; record owner and chain; hand off; update; recover; transfer; wipe; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show device, hardware, firmware, identity reference, owner, configuration, action, evidence, and limitation without exposing secret material. Require human approval for key, assignment, wipe, and retirement.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use secure provisioning station, unique identity, protected keys, signed firmware, least privilege, inventory, policy, enrollment, audit, recovery, and secure handoff.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show device identity, owner, version, posture, configuration, enrollment, evidence freshness, and authorized assign, restrict, update, recover, transfer, wipe, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Provisioning charter; device and chain-of-custody; identity and key metadata; firmware and configuration; validation, handoff, recovery, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify device, hardware, firmware, identity reference, station, operator, owner, method, time, result, and limitation. Enrollment proves only the recorded action, not complete trust.

VERIFICATION AND ACCEPTANCE TESTS

Verify identity, keys, firmware, configuration, ownership, enrollment, network, security, privacy, support, update, recovery, transfer, wipe, and inventory.

FAILURE STATES AND ANTI-PATTERNS

Shared identity, key exposure, wrong owner, firmware mismatch, enrollment without validation, chain break, wipe without scope, and device status treated as security.

RECOVERY AND RECONSIDERATION

Quarantine, preserve evidence, revoke identity, rotate keys, restore known-good firmware, re-provision, notify owner, and reassess fleet or user impact. Reopen on hardware, firmware, owner, policy, or threat change.

COMPLETION CRITERIA

Device, owner, identity, firmware, keys, configuration, enrollment, validation, evidence, recovery, wipe, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve device metadata, provisioning, ownership, keys references, firmware, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Manufacturing, Embedded, Security, Mobile, Operations, Support, and Disposal.

RELATED CODEX SECTIONS

198-199 Embedded and Hardware Security; 204-205 Keys and Secrets; 258 Configuration; 294 Provisioning; 354 Certification; 356 Logistics; 399 Hardware Disposal.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, device, hardware, identity, security, manufacturing, and operations context, and the integration roadmap. Exact devices, owners, stations, keys, firmware, privacy, wipe, and acceptance authority remain unresolved.

356

SECTION 356 - LOGISTICS

CODEX/03_SECTIONS/356_LOGISTICS.txt

SHA-256 d0f4c6cd0543c6bfd2072f7796832c4e8d8d093dde29ad2c90c7aab32bb39839

4,793 BYTES / PRESERVED SOURCE IDENTITY

SECTION 356 — Logistics

STATUS: CONTENT-COMPLETE

PURPOSE

Move materials, products, devices, documents, data, and equipment safely, traceably, timely, securely, and cost-effectively across locations and custody boundaries.

REQUIRED OUTCOME

A logistics model with item identity, packaging, custody, route, carrier, condition, timing, security, customs or jurisdiction, exception, receipt, damage, and recovery.

WHY THIS MATTERS

Movement introduces loss, tampering, delay, damage, misrouting, privacy, safety, and chain-of-custody risk. Shipment status is not proof of condition, receipt, or correct destination.

APPLICABILITY

Apply when physical or digital items are transported, staged, stored in transit, delivered, returned, or transferred between custodians.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no item, custody, route, delivery, storage-in-transit, or transport risk changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use supply chain, manufacturing, device, packaging, distribution, security, privacy, legal, customs, inventory, and customer baselines.

REQUIRED INPUTS

Item, identity, quantity, condition, packaging, origin, destination, custodian, carrier, route, timing, security, receipt, and exception.

HUMAN RESPONSIBILITIES AND AUTHORITY

Logistics owners approve route, carrier, custody, security, condition, cost, exception, and loss response. Humans verify receipt and resolve damage or claims.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized shipment metadata, forecast delay, draft routes, and identify anomalies. It must not expose customer data, change destination, certify condition, or accept delivery without authority.

APPLICABLE EXISTING AGENT ROLES

Logistics Manager, Supply Chain Manager, Device Provisioning Engineer, Security Auditor, Procurement Analyst, Analyst, Observer, Verifier, and Coordinator.

REQUIRED ACTIVITIES

Identify and package; assign custody; select route and carrier; ship; track; handle customs or jurisdiction; receive and inspect; reconcile inventory; address exception; return or dispose.

HUMAN-AI INTERACTION REQUIREMENTS

Show item, custody, route, destination, condition, evidence, action, privacy, and limitation. Require human approval for high-value, sensitive, regulated, hazardous, or disputed shipment.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use serialization, chain of custody, tamper evidence, condition, tracking, access, privacy, route resilience, inventory reconciliation, and exception workflow.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show shipment, item, custody, location, condition, ETA, exception, evidence freshness, owner, and authorized hold, reroute, inspect, accept, return, and claim controls.

REQUIRED OUTPUTS AND ARTIFACTS

Logistics charter; item and custody; packaging and route; carrier and tracking; receipt and exception; decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify item, serial or version, quantity, custodian, origin, destination, carrier, method, time, condition, result, and limitation. Tracking event proves location signal, not condition or receipt.

VERIFICATION AND ACCEPTANCE TESTS

Verify identity, quantity, packaging, custody, route, condition, tracking, receipt, security, privacy, customs, damage, return, and inventory reconciliation.

FAILURE STATES AND ANTI-PATTERNS

Untracked transfer, wrong destination, custody gap, tamper ignored, sensitive data exposed, receipt assumed, carrier single point, and delivery speed over safety or condition.

RECOVERY AND RECONSIDERATION

Hold, investigate, preserve custody evidence, reroute or replace, notify owners and recipients, reconcile inventory, and reassess security and customer impact. Reopen on route, carrier, item, condition, or jurisdiction change.

COMPLETION CRITERIA

Items, custody, routes, carriers, condition, evidence, owners, exceptions, receipt, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve shipment, custody, tracking, receipt, damage, claims, decisions, evidence, exceptions, and open gaps. Hand off to Distribution, Manufacturing, Device Provisioning, Vendor, Support, Finance, and Operations.

RELATED CODEX SECTIONS

350 Supply Chain; 351 Manufacturing; 355 Device Provisioning; 357 Distribution; 399 Hardware Disposal; 348 Procurement; 334 Customer Support.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, supply, manufacturing, device, security, customer, and operations context, and the integration roadmap. Exact items, routes, carriers, jurisdictions, custody, condition, and acceptance authority remain unresolved.

357

SECTION 357 - DISTRIBUTION

CODEX/03_SECTIONS/357_DISTRIBUTION.txt

SHA-256 851b0a6d603c7685c5ff2a4bc3e1d7ab6c247ca749105f33f0483002c84090c1

4,747 BYTES / PRESERVED SOURCE IDENTITY

SECTION 357 — Distribution

STATUS: CONTENT-COMPLETE

PURPOSE

Deliver products, devices, packages, content, data, or services to authorized customers, channels, locations, or systems with identity, condition, entitlement, evidence, and support.

REQUIRED OUTCOME

A distribution model with item or artifact, audience, channel, inventory, entitlement, route, condition, receipt, support, withdrawal, and reconciliation.

WHY THIS MATTERS

Distribution turns inventory into customer or operational state. Incorrect version, quantity, audience, location, or condition can create financial, legal, safety, and trust consequences.

APPLICABILITY

Apply when goods, software, firmware, data, content, or services are delivered to a customer, partner, site, device, or downstream system.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no delivery, audience, item, entitlement, channel, receipt, or customer outcome changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use logistics, packaging, release, licensing, commerce, customer, inventory, support, security, and legal baselines.

REQUIRED INPUTS

Item or artifact, version, quantity, customer, entitlement, channel, route, condition, inventory, receipt, support, and withdrawal.

HUMAN RESPONSIBILITIES AND AUTHORITY

Distribution owners approve audience, entitlement, channel, route, condition, customer commitment, exception, withdrawal, and claim. Humans resolve disputes.

AI RESPONSIBILITIES AND BOUNDARIES

AI may validate authorized order and inventory metadata, draft allocation, and identify anomalies. It must not release, change quantity, widen audience, expose data, or certify receipt without authority.

APPLICABLE EXISTING AGENT ROLES

Distribution Manager, Logistics Manager, Release Engineer, Commerce Analyst, Customer Support, Security Auditor, Analyst, Verifier, and Coordinator.

REQUIRED ACTIVITIES

Validate order and entitlement; allocate; package; route; distribute; track; confirm receipt and condition; support; withdraw or replace; reconcile inventory and finance; and record.

HUMAN-AI INTERACTION REQUIREMENTS

Show item, version, customer, entitlement, channel, route, condition, evidence, and limitation. Require human approval for sensitive, regulated, high-value, disputed, or irreversible distribution.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use order-to-delivery identity, entitlement, inventory, provenance, tamper or integrity, receipt, withdrawal, support, and reconciliation controls.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show distribution state, item, customer, entitlement, route, receipt, condition, evidence freshness, owner, and authorized hold, release, reroute, withdraw, replace, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Distribution charter; order and entitlement; item and route; receipt and condition; support and withdrawal; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify item, version, customer, entitlement, channel, route, method, time, receipt, condition, actor, and limitation. Shipment or download proves only delivery event.

VERIFICATION AND ACCEPTANCE TESTS

Verify identity, quantity, version, entitlement, route, condition, receipt, support, withdrawal, refund or replacement, inventory, and financial reconciliation.

FAILURE STATES AND ANTI-PATTERNS

Wrong customer or version, entitlement bypass, receipt assumed, inventory drift, withdrawal impossible, condition unverified, and delivery metric treated as customer success.

RECOVERY AND RECONSIDERATION

Hold or withdraw, preserve evidence, notify recipients, replace or refund, reconcile inventory and finance, and reassess impact. Reopen on item, customer, channel, route, or entitlement change.

COMPLETION CRITERIA

Item, customer, entitlement, route, condition, receipt, support, evidence, owners, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve distribution, logistics, orders, receipts, support, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Commerce, Billing, Customer Support, Logistics, Release, and Operations.

RELATED CODEX SECTIONS

291 Distribution Engineering; 356 Logistics; 358 Commerce; 359-362 Billing, Payments, Subscriptions, and Revenue; 334 Customer Support; 391-400 Lifecycle.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, distribution, logistics, commerce, customer, finance, and support context, and the integration roadmap. Exact products, channels, entitlements, jurisdictions, receipts, refunds, and acceptance authority remain unresolved.

358

SECTION 358 - COMMERCE

CODEX/03_SECTIONS/358_COMMERCE.txt

SHA-256 6ac4bc55b00523008adc5d78be16ea514bc15fac0416e9930cbd60d7f8b42fdd

5,047 BYTES / PRESERVED SOURCE IDENTITY

SECTION 358 — Commerce

STATUS: CONTENT-COMPLETE

PURPOSE

Govern the customer transaction lifecycle from offer, catalog, price, quote, order, entitlement, fulfillment, support, refund, dispute, and records.

REQUIRED OUTCOME

A commerce model with product, customer, price, tax or jurisdiction, offer, quote, order, entitlement, fulfillment, payment dependency, cancellation, refund, evidence, and ownership.

WHY THIS MATTERS

Commerce joins customer trust, money, product truth, legal terms, fulfillment, and accounting. A technically successful transaction can still be wrong, unauthorized, noncompliant, or impossible to unwind.

APPLICABILITY

Apply when a product or service is offered, priced, quoted, ordered, entitled, fulfilled, renewed, refunded, disputed, or retired commercially.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no commercial offer, customer, price, order, entitlement, fulfillment, or financial obligation changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use product, pricing, legal, tax, privacy, identity, payments, billing, subscription, distribution, support, and records baselines.

REQUIRED INPUTS

Product, customer, catalog, price, offer, terms, jurisdiction, quote, order, entitlement, fulfillment, payment, refund, dispute, and evidence.

HUMAN RESPONSIBILITIES AND AUTHORITY

Commerce owners approve product and price, terms, discounts, customer, jurisdiction, exception, refund, dispute, and retirement. Finance and legal owners approve their obligations.

AI RESPONSIBILITIES AND BOUNDARIES

AI may summarize authorized catalog and order data, draft quote analysis, and identify inconsistency. It must not set price, make offer, infer customer eligibility, alter order, approve refund, or expose financial data without authority.

APPLICABLE EXISTING AGENT ROLES

Commerce Manager, Product Manager, Financial Analyst, Legal Advisor, Payments Engineer, Billing Engineer, Customer Support, Analyst, Verifier, and Governance Reviewer.

REQUIRED ACTIVITIES

Define catalog and offer; price and quote; validate customer and terms; order; authorize; fulfill; bill and pay; support; refund or dispute; reconcile; report; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show product, customer, price, terms, jurisdiction, order, entitlement, evidence, authority, and limitation. Require human approval for discount, high-value, exception, refund, dispute, and legal or tax action.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link catalog, price, terms, customer, order, entitlement, payment, fulfillment, billing, refund, and ledger with immutable history, idempotence, authorization, and reconciliation.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show catalog, order, customer, entitlement, price, payment, fulfillment, disputes, evidence freshness, owner, and authorized approve, hold, refund, cancel, correct, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Commerce charter; catalog and pricing; offer and order; entitlement and fulfillment; billing, payment, refund, dispute, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify product, customer, price, terms, jurisdiction, order, actor, method, time, result, entitlement, and limitation. Order or payment status proves only its recorded transaction state.

VERIFICATION AND ACCEPTANCE TESTS

Verify product, price, terms, customer, entitlement, authorization, fulfillment, payment, tax, refund, dispute, reconciliation, privacy, and support. Test duplicate, partial, failed, and reversed transaction.

FAILURE STATES AND ANTI-PATTERNS

Price drift, entitlement without payment, payment without fulfillment, duplicate order, refund without ledger, hidden terms, wrong jurisdiction, and transaction success treated as customer outcome.

RECOVERY AND RECONSIDERATION

Hold, reverse or refund, preserve records, reconcile, communicate, support customer, and reassess legal and financial impact. Reopen on product, price, terms, jurisdiction, payment, or fulfillment change.

COMPLETION CRITERIA

Product, customer, price, terms, order, entitlement, fulfillment, payment, billing, evidence, owners, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve catalog, prices, orders, entitlements, payments, billing, refunds, disputes, decisions, evidence, exceptions, and open gaps. Hand off to Billing, Payments, Subscriptions, Revenue, Support, Legal, Finance, and Operations.

RELATED CODEX SECTIONS

219 Legal; 348 Procurement; 357 Distribution; 359-362 Billing, Payments, Subscriptions, and Revenue Operations; 334 Customer Support; 225 Records.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, product, commerce, financial, legal, payment, customer, and evidence context, and the integration roadmap. Exact products, prices, jurisdictions, tax, terms, systems, refunds, and acceptance authority remain unresolved.

359

SECTION 359 - BILLING ENGINEERING

CODEX/03_SECTIONS/359_BILLING_ENGINEERING.txt

SHA-256 d77dd0ebf6f91af1a59af2cab6c2506c77d51eea97a20943c1956f6d55dbf9bb

4,867 BYTES / PRESERVED SOURCE IDENTITY

SECTION 359 — Billing Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Calculate, present, collect, adjust, reconcile, and record charges accurately, transparently, securely, and according to authorized product, contract, tax, and customer terms.

REQUIRED OUTCOME

A billing model with customer, product, price, usage, period, currency, tax, invoice, adjustment, payment dependency, ledger, dispute, audit, and correction.

WHY THIS MATTERS

Billing errors affect money, trust, legal obligations, service access, and records. A generated invoice or successful charge does not prove correct entitlement, price, usage, tax, or customer impact.

APPLICABILITY

Apply when charges, invoices, usage, credits, taxes, adjustments, collections, or financial records are calculated or changed.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no billing, price, usage, customer, invoice, adjustment, or financial obligation changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use commerce, pricing, contracts, payments, subscriptions, usage metering, finance, tax, legal, privacy, identity, and records baselines.

REQUIRED INPUTS

Customer, product, price, usage, period, currency, tax, terms, invoice, ledger, adjustments, payment, dispute, and evidence.

HUMAN RESPONSIBILITIES AND AUTHORITY

Billing owners approve price and rule, invoice, adjustment, credit, write-off, dispute, access, and correction. Finance and legal owners approve material accounting or tax decisions.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized billing records, identify anomalies, and draft reconciliation or customer explanation. It must not expose payment data, change charges, issue credit, or close disputes without authority.

APPLICABLE EXISTING AGENT ROLES

Billing Engineer, Financial Analyst, Commerce Manager, Payments Engineer, Subscription Manager, Legal Advisor, Data Analyst, Tester, Verifier, and Governance Reviewer.

REQUIRED ACTIVITIES

Define pricing and usage; calculate; validate; invoice; collect; reconcile; adjust; handle dispute; report; secure records; correct; and close or retire billing rule.

HUMAN-AI INTERACTION REQUIREMENTS

Show customer, product, usage, period, price, currency, tax, rule, invoice, evidence, authority, and limitation. Require human approval for charge, adjustment, credit, write-off, dispute, and disclosure.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use immutable invoice and ledger history, idempotent calculation, usage lineage, versioned price, access segregation, reconciliation, audit, correction, and customer explanation.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show invoice, usage, price, period, payment, dispute, evidence freshness, owner, and authorized hold, correct, credit, refund, reconcile, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Billing charter; price and usage; calculation and invoice; ledger and reconciliation; adjustment and dispute; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify customer, product, usage, price version, period, currency, tax, rule, method, time, result, actor, and limitation. Invoice generation proves only calculation execution.

VERIFICATION AND ACCEPTANCE TESTS

Verify price, usage, period, tax, currency, entitlement, invoice, ledger, payment, adjustment, refund, dispute, reconciliation, privacy, and audit. Test duplicate, late, corrected, and missing usage.

FAILURE STATES AND ANTI-PATTERNS

Price drift, hidden usage, duplicate charge, invoice-ledger mismatch, credit without authority, tax assumption, payment data exposure, and billing status used as product truth.

RECOVERY AND RECONSIDERATION

Hold collection, preserve ledger, correct or reverse, refund or credit, communicate, reconcile, and reassess access and legal impact. Reopen on price, usage, contract, tax, payment, or customer dispute.

COMPLETION CRITERIA

Customer, product, price, usage, period, invoice, ledger, tax, payment, evidence, owners, corrections, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve billing rules, invoices, usage, ledgers, adjustments, disputes, decisions, evidence, exceptions, and open gaps. Hand off to Payments, Commerce, Subscriptions, Revenue, Finance, Legal, Support, and Records.

RELATED CODEX SECTIONS

358 Commerce; 360 Payments; 361 Subscriptions; 362 Revenue Operations; 365 Usage Metering; 359 Billing; 225 Records; 226 Audit.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, commerce, finance, payment, subscription, legal, data, and evidence context, and the integration roadmap. Exact pricing, tax, currencies, rules, ledgers, jurisdictions, disputes, and acceptance authority remain unresolved.

360

SECTION 360 - PAYMENTS ENGINEERING

CODEX/03_SECTIONS/360_PAYMENTS_ENGINEERING.txt

SHA-256 bd4dca904a022d6e5324c8c207af58bedc75e956e1967aece9acc7aeeac49605

4,958 BYTES / PRESERVED SOURCE IDENTITY

SECTION 360 — Payments Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Authorize, initiate, authenticate, settle, reconcile, reverse, refund, and record payments securely, reliably, transparently, and within customer and regulatory boundaries.

REQUIRED OUTCOME

A payments model with instrument, customer, authorization, provider, amount, currency, idempotency, status, fraud, settlement, dispute, refund, privacy, and reconciliation.

WHY THIS MATTERS

Payments combine money, identity, availability, fraud, privacy, and legal obligations. A provider response can be delayed, duplicated, reversed, or inconsistent with commerce and ledger state.

APPLICABILITY

Apply when a system initiates, accepts, authorizes, settles, refunds, disputes, or reconciles monetary payment.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no payment, customer, amount, instrument, provider, settlement, refund, or financial record changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use commerce, billing, identity, fraud, privacy, security, legal, provider, ledger, records, and reconciliation baselines.

REQUIRED INPUTS

Customer, amount, currency, product, instrument reference, provider, authorization, idempotency, status, settlement, refund, dispute, and evidence.

HUMAN RESPONSIBILITIES AND AUTHORITY

Payments owners approve provider, instrument, limits, fraud, refund, dispute, access, exception, and reconciliation. Humans handle high-risk or disputed financial decisions.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized payment metadata, identify anomalies, and draft reconciliation or support explanations. It must not access instrument secrets, initiate or reverse payment, approve refund, or decide fraud without authority.

APPLICABLE EXISTING AGENT ROLES

Payments Engineer, Billing Engineer, Fraud Analyst, Security Auditor, Privacy Reviewer, Commerce Manager, Financial Analyst, Tester, Verifier, and Incident Responder.

REQUIRED ACTIVITIES

Define payment state; tokenize or reference; authorize; handle provider response; enforce idempotence; settle; reconcile; refund or dispute; monitor; secure; and recover.

HUMAN-AI INTERACTION REQUIREMENTS

Show payment reference, customer, amount, currency, provider, status, action, authority, evidence, and limitation without exposing instrument data. Require human approval for refund, reversal, dispute, and exception.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use tokenization, least privilege, idempotency, state machine, provider abstraction, ledger, webhook validation, audit, privacy, fraud control, and retry without duplicate charge.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show payment state, provider, amount, currency, settlement, dispute, evidence freshness, owner, and authorized hold, retry, refund, reverse, reconcile, and restrict controls.

REQUIRED OUTPUTS AND ARTIFACTS

Payments charter; state and provider; instrument and security; authorization and settlement; reconciliation, refund, dispute, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify payment reference, customer, amount, currency, provider, state, method, time, actor, result, and limitation. Provider response proves only its recorded response and status.

VERIFICATION AND ACCEPTANCE TESTS

Verify authorization, token, amount, currency, idempotence, provider, webhook, settlement, refund, dispute, ledger, privacy, fraud, and recovery. Test timeout, duplicate, reversal, and provider outage.

FAILURE STATES AND ANTI-PATTERNS

Instrument data in logs, duplicate charge, provider state as ledger truth, retry without idempotence, refund without authorization, webhook trust, and payment success treated as fulfillment.

RECOVERY AND RECONSIDERATION

Freeze or reconcile, preserve provider and ledger evidence, refund or reverse through authority, communicate, protect customer, and reassess. Reopen on provider, state, fraud, contract, or dispute change.

COMPLETION CRITERIA

Customer, amount, currency, provider, state, authorization, settlement, refund, dispute, evidence, owners, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve references, states, provider events, ledgers, disputes, decisions, evidence, exceptions, and open gaps. Hand off to Billing, Commerce, Fraud, Finance, Legal, Support, Security, and Records.

RELATED CODEX SECTIONS

358 Commerce; 359 Billing; 361 Subscription Management; 215 Fraud Prevention; 200-202 IAM; 218 Privacy; 226 Audit; 225 Records.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, commerce, finance, payments, security, privacy, fraud, and evidence context, and the integration roadmap. Exact providers, instruments, currencies, jurisdictions, fraud, settlement, refunds, and acceptance authority remain unresolved.

361

SECTION 361 - SUBSCRIPTION MANAGEMENT

CODEX/03_SECTIONS/361_SUBSCRIPTION_MANAGEMENT.txt

SHA-256 ea63337409243a0e4832400cd634dc4fe8d57291fd39ad3ba445677b97a1e75c

4,729 BYTES / PRESERVED SOURCE IDENTITY

SECTION 361 — Subscription Management

STATUS: CONTENT-COMPLETE

PURPOSE

Manage recurring entitlement, plan, term, renewal, upgrade, downgrade, pause, cancellation, billing, payment, access, and customer communication accurately and fairly.

REQUIRED OUTCOME

A subscription model with customer, plan, term, entitlement, price, usage, renewal, notice, payment, status, cancellation, refund, support, evidence, and lifecycle.

WHY THIS MATTERS

Subscription state controls access and recurring financial obligation. Ambiguous renewal, failed payment, entitlements, or cancellation can create customer harm and legal exposure.

APPLICABILITY

Apply when a product or service grants recurring or term-based access, billing, usage, or commitment.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no recurring entitlement, term, price, renewal, cancellation, payment, or customer obligation changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use commerce, billing, payments, identity, product, legal, privacy, support, usage, and records baselines.

REQUIRED INPUTS

Customer, plan, price, term, entitlement, usage, renewal, notice, payment, status, cancellation, refund, support, and evidence.

HUMAN RESPONSIBILITIES AND AUTHORITY

Subscription owners approve plans, price, terms, renewal, grace, cancellation, refund, access, exception, and customer communication. Humans resolve disputes and legal obligations.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized subscription state, identify anomalies, and draft customer explanations. It must not alter entitlement, renew, cancel, charge, refund, or disclose customer data without authority.

APPLICABLE EXISTING AGENT ROLES

Subscription Manager, Commerce Manager, Billing Engineer, Payments Engineer, Customer Support, Legal Advisor, Analyst, Tester, Verifier, and Governance Reviewer.

REQUIRED ACTIVITIES

Define plan and entitlement; create; activate; bill; renew; handle payment failure; upgrade or downgrade; pause; cancel; refund; notify; reconcile; and retire plan.

HUMAN-AI INTERACTION REQUIREMENTS

Show customer, plan, term, price, entitlement, status, payment, action, evidence, authority, and limitation. Require human approval for refund, exception, disputed cancellation, and high-impact access.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use explicit state machine, immutable history, idempotence, entitlement and billing linkage, notice, grace, access, privacy, audit, and reconciliation.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show subscription, plan, term, renewal, entitlement, payment, cancellation, dispute, evidence freshness, owner, and authorized change, pause, cancel, refund, restore, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Subscription charter; plan and entitlement; term and price; lifecycle and payment; notice, cancellation, refund, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify customer, subscription, plan, term, price, entitlement, payment, state, notice, method, time, result, and limitation. State transition proves only recorded system state.

VERIFICATION AND ACCEPTANCE TESTS

Verify create, activate, renew, grace, payment failure, upgrade, downgrade, pause, cancel, refund, entitlement, notice, privacy, support, and reconciliation.

FAILURE STATES AND ANTI-PATTERNS

Unclear renewal, access after cancel, charge after failure, cancellation maze, entitlement mismatch, hidden term, refund without ledger, and subscription count as value.

RECOVERY AND RECONSIDERATION

Correct state, stop charge, restore or restrict access, refund or credit, communicate, preserve evidence, and reassess customer and legal impact. Reopen on plan, price, term, payment, or policy change.

COMPLETION CRITERIA

Customer, plan, term, price, entitlement, payment, lifecycle, notices, evidence, owners, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve subscription state, plans, notices, payments, billing, cancellations, disputes, decisions, evidence, exceptions, and open gaps. Hand off to Commerce, Billing, Payments, Support, Legal, Revenue, and Records.

RELATED CODEX SECTIONS

358 Commerce; 359 Billing; 360 Payments; 362 Revenue Operations; 334 Customer Support; 219 Legal; 225 Records.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, commerce, finance, payments, customer, legal, privacy, and service context, and the integration roadmap. Exact plans, terms, prices, jurisdictions, notices, entitlements, refunds, and acceptance authority remain unresolved.

362

SECTION 362 - REVENUE OPERATIONS

CODEX/03_SECTIONS/362_REVENUE_OPERATIONS.txt

SHA-256 dc9d74b86f7028f43d95529c998e2f57d30d57ab93fcc261fa13ac5fcaf514ac

5,272 BYTES / PRESERVED SOURCE IDENTITY

SECTION 362 — Revenue Operations

STATUS: CONTENT-COMPLETE

PURPOSE

Coordinate the data, process, systems, people, controls, forecasting, pipeline, billing, customer, and finance workflows that produce trustworthy revenue operations.

REQUIRED OUTCOME

A revenue-operations model with lifecycle, source systems, customer and product data, pipeline, pricing, quote, order, billing, payment, subscription, forecast, reconciliation, controls, and ownership.

WHY THIS MATTERS

Revenue data crosses teams and systems with different definitions and incentives. A polished forecast or dashboard can hide timing, attribution, churn, billing, data, or recognition error.

APPLICABILITY

Apply when commercial process, customer lifecycle, revenue data, forecasting, billing, payments, subscriptions, sales operations, or financial reporting is designed or operated.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no commercial lifecycle, revenue data, financial process, customer, forecast, or control behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use commerce, product, sales, billing, payments, subscriptions, finance, legal, data, privacy, audit, controls, and service baselines.

REQUIRED INPUTS

Customers, products, prices, pipeline, quotes, orders, usage, subscriptions, invoices, payments, adjustments, forecast, definitions, controls, and evidence.

HUMAN RESPONSIBILITIES AND AUTHORITY

Revenue owners define lifecycle, definitions, controls, forecast, approvals, exceptions, reconciliation, and reporting. Finance, legal, sales, product, and customer owners approve their domains.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized commercial data, identify anomalies, draft forecasts and process improvements, and summarize evidence. It must not alter financial records, make commitments, infer customer intent, approve transactions, or attest without authority.

APPLICABLE EXISTING AGENT ROLES

Revenue Operations Manager, Financial Analyst, Commerce Manager, Billing Engineer, Payments Engineer, Data Analyst, Sales Engineer, Auditor, Verifier, and Governance Reviewer.

REQUIRED ACTIVITIES

Define lifecycle and terms; align systems and definitions; manage pipeline and quote; order and entitlement; bill and collect; reconcile; forecast; report; control access; investigate; and improve.

HUMAN-AI INTERACTION REQUIREMENTS

Show customer, product, lifecycle, source, definition, period, forecast, action, authority, evidence, and limitation. Require human approval for forecast, financial adjustment, commitment, report, and exception.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link CRM, product, commerce, billing, payment, subscription, usage, ledger, support, and finance with identity, lineage, immutable history, reconciliation, and separation of duties.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show pipeline, customer, product, order, subscription, invoice, payment, forecast, reconciliation, evidence freshness, owner, and authorized correct, approve, hold, reconcile, report, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Revenue-operations charter; lifecycle and system map; definitions and controls; pipeline, forecast, billing, payment, subscription, reconciliation, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify customer, product, transaction, source, definition, period, method, result, actor, forecast, and limitation. Forecast or revenue report proves only its source, period, and assumptions.

VERIFICATION AND ACCEPTANCE TESTS

Verify lifecycle, definitions, access, quote, order, entitlement, billing, payment, subscription, usage, forecast, reconciliation, controls, audit, privacy, and corrections.

FAILURE STATES AND ANTI-PATTERNS

Conflicting definitions, forecast as fact, pipeline inflation, revenue dashboard without lineage, manual adjustment hidden, customer data overexposure, and metric gaming.

RECOVERY AND RECONSIDERATION

Freeze affected report or adjustment, preserve records, reconcile, correct, communicate, and reassess financial, customer, legal, and operational impact. Reopen on source, definition, price, product, contract, or period change.

COMPLETION CRITERIA

Lifecycle, systems, definitions, customers, products, transactions, controls, owners, evidence, reconciliation, forecasts, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve lifecycle data, definitions, transactions, reports, forecasts, adjustments, decisions, evidence, exceptions, and open gaps. Hand off to Commerce, Billing, Payments, Subscriptions, Finance, Sales, Product, Audit, and Records.

RELATED CODEX SECTIONS

010 Product Management; 343 Product Marketing; 346-347 Sales and Solutions; 358-361 Commerce, Billing, Payments, and Subscriptions; 365 Usage Metering; 226 Audit; 227 Controls.

CODEX/03_SECTIONS/362_REVENUE_OPERATIONS.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, commerce, finance, data, audit, customer, and control context, and the integration roadmap. Exact lifecycle, definitions, systems, period, recognition, forecasts, controls, jurisdictions, and acceptance authority remain unresolved.

363

SECTION 363 - FINANCIAL OPERATIONS

CODEX/03_SECTIONS/363_FINANCIAL_OPERATIONS.txt

SHA-256 3076127fa0e369d22f903c2ee0bee2fa9f68ef465bab25ebabb304ead98f3517

4,918 BYTES / PRESERVED SOURCE IDENTITY

SECTION 363 — Financial Operations

STATUS: CONTENT-COMPLETE

PURPOSE

Record, control, reconcile, forecast, report, and protect financial activity, resources, obligations, approvals, and evidence across the lifecycle.

REQUIRED OUTCOME

A financial-operations model with accounts, transactions, budgets, commitments, controls, approvals, reconciliation, reporting, close, audit, privacy, and correction.

WHY THIS MATTERS

Financial records drive decisions, obligations, trust, and legal reporting. A dashboard or ledger entry can be wrong through timing, classification, duplicate, omission, adjustment, or unsupported assumption.

APPLICABILITY

Apply when money, assets, commitments, budgets, costs, revenue, expenses, payments, forecasts, or financial reports are created, changed, or controlled.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no financial value, transaction, budget, obligation, control, or report changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use commerce, billing, payments, procurement, controls, audit, legal, tax, records, data, and policy baselines.

REQUIRED INPUTS

Accounts, transactions, invoices, payments, budgets, commitments, prices, evidence, approvals, controls, reconciliations, periods, and reports.

HUMAN RESPONSIBILITIES AND AUTHORITY

Finance owners approve accounting meaning, budget, adjustment, payment, write-off, close, report, access, exception, and attestation. Humans preserve segregation of duties and resolve disputes.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized financial metadata, identify anomalies, draft reconciliations, and summarize reports. It must not alter ledger, approve payment, set accounting treatment, disclose data, or attest without human authority.

APPLICABLE EXISTING AGENT ROLES

Financial Analyst, Accountant, Auditor, Controls Analyst, Commerce Manager, Billing Engineer, Payments Engineer, Researcher, Verifier, and Governance Reviewer.

REQUIRED ACTIVITIES

Define accounts and periods; record; approve; reconcile; budget; forecast; close; report; control access; investigate variance; correct; audit; and preserve or retire records.

HUMAN-AI INTERACTION REQUIREMENTS

Show account, transaction, period, source, treatment, approval, evidence, variance, and limitation. Require human review for adjustment, close, report, attestation, payment, and exception.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use immutable ledger history, segregation, approval, reconciliation, period close, lineage, access, retention, correction, and audit. Separate operational estimates from authoritative records.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show period, balances, transactions, budget, variance, approvals, evidence freshness, owner, and authorized hold, adjust, reconcile, close, report, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Financial-operations charter; account and control; transaction and reconciliation; budget and forecast; close, report, audit, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify account, transaction, period, source, treatment, method, time, actor, approval, result, and limitation. Report or ledger status proves only recorded state and scope.

VERIFICATION AND ACCEPTANCE TESTS

Verify authorization, segregation, completeness, accuracy, cutoff, reconciliation, adjustment, period close, report, access, retention, and audit. Test duplicate, late, reversed, and corrected transaction.

FAILURE STATES AND ANTI-PATTERNS

Unreconciled ledger, unsupported adjustment, shared approval, period drift, forecast as fact, report without source, and financial control as paperwork only.

RECOVERY AND RECONSIDERATION

Freeze affected record or period, preserve evidence, correct or reverse, reconcile, communicate, and reassess legal and financial impact. Reopen on transaction, rule, period, source, or obligation change.

COMPLETION CRITERIA

Accounts, transactions, periods, controls, approvals, reconciliation, reports, evidence, owners, corrections, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending authorized acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve ledger, budgets, reports, reconciliations, approvals, adjustments, decisions, evidence, exceptions, and open gaps. Hand off to Billing, Payments, Commerce, Revenue, Audit, Legal, Procurement, and Records.

RELATED CODEX SECTIONS

226 Audit; 227 Internal Controls; 348 Procurement; 358-362 Commerce, Billing, Payments, Subscriptions, and Revenue; 364 Licensing; 225 Records.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, finance, commerce, audit, controls, legal, and evidence context, and the integration roadmap. Exact accounts, standards, periods, tax, authority, reconciliation, and acceptance evidence remain unresolved.

364

SECTION 364 - LICENSING OPERATIONS

CODEX/03_SECTIONS/364_LICENSING_OPERATIONS.txt

SHA-256 5c664dd4db3f71f8c7903ff280faf91e5dd5dd9b9b1bded71f4a1653e5fa025c

4,789 BYTES / PRESERVED SOURCE IDENTITY

SECTION 364 — Licensing Operations

STATUS: CONTENT-COMPLETE

PURPOSE

Operate licenses, entitlements, seats, terms, renewals, usage, restrictions, notices, compliance, and termination across products, software, data, content, and services.

REQUIRED OUTCOME

A licensing-operations model with asset, license, owner, customer, entitlement, term, use, usage evidence, renewal, restriction, notice, audit, and exit.

WHY THIS MATTERS

License rights can expire, narrow, or depend on use and distribution. A license record or purchase does not prove current entitlement, compliant use, or correct customer access.

APPLICABILITY

Apply when licensed material or entitlement is activated, measured, assigned, renewed, transferred, audited, restricted, or retired.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no licensed asset, entitlement, use, term, renewal, distribution, or restriction changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use IP, legal, procurement, commerce, subscription, package, dependency, asset, finance, audit, and records baselines.

REQUIRED INPUTS

Asset, license, licensor, customer, entitlement, seats, term, usage, restriction, notice, renewal, audit, transfer, and termination.

HUMAN RESPONSIBILITIES AND AUTHORITY

License owners and counsel approve use, entitlement, distribution, renewal, exception, audit response, termination, and customer restriction. Humans resolve ambiguous terms.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized license metadata and usage, identify mismatch, and draft renewal or remediation. It must not interpret ambiguous terms conclusively, revoke access, disclose licensed content, or approve use without human authority.

APPLICABLE EXISTING AGENT ROLES

License Manager, Legal Advisor, IP Analyst, Procurement Analyst, Commerce Manager, Auditor, Researcher, Verifier, and Governance Reviewer.

REQUIRED ACTIVITIES

Inventory licenses; assign owners; map assets and customers; provision entitlement; measure use; monitor term and restrictions; renew; remediate; audit; revoke or transfer; and terminate.

HUMAN-AI INTERACTION REQUIREMENTS

Show asset, license, customer, term, entitlement, use, restriction, evidence, and limitation. Require human or counsel review for ambiguous, high-impact, renewal, audit, and termination decisions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link asset, license, customer, entitlement, usage, billing, distribution, access, notice, audit, and termination. Prevent license state from silently diverging from service access.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show licenses, assets, terms, entitlements, use, expiry, customers, evidence freshness, owner, and authorized assign, renew, restrict, transfer, audit, and terminate controls.

REQUIRED OUTPUTS AND ARTIFACTS

Licensing-operations charter; asset and license; entitlement and usage; renewal and restriction; audit, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify asset, license, licensor, customer, term, entitlement, usage, source, method, time, result, and limitation. Entitlement record proves only its scope and time.

VERIFICATION AND ACCEPTANCE TESTS

Verify license, asset, customer, entitlement, term, usage, restriction, billing, notice, renewal, transfer, audit, revocation, and termination. Test expiry and excess use.

FAILURE STATES AND ANTI-PATTERNS

Expired license, entitlement mismatch, usage unmeasured, restriction hidden, audit evidence missing, rights assumed from purchase, and revocation without customer process.

RECOVERY AND RECONSIDERATION

Restrict or cure use, notify owners and customers, renew or replace, preserve evidence, correct billing and access, and reassess downstream artifacts. Reopen on license, asset, customer, term, or law change.

COMPLETION CRITERIA

Assets, licenses, owners, customers, entitlements, terms, usage, evidence, restrictions, renewal, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve licenses, entitlements, usage, renewals, audits, decisions, evidence, exceptions, and open gaps. Hand off to IP, Legal, Procurement, Commerce, Subscriptions, Billing, Distribution, and Records.

RELATED CODEX SECTIONS

223-224 IP and Licensing; 348 Procurement; 358 Commerce; 361 Subscriptions; 359 Billing; 291 and 357 Distribution; 225 Records.

CODEX/03_SECTIONS/364_LICENSING_OPERATIONS.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, licensing, legal, commerce, finance, and records context, and the integration roadmap. Exact assets, licenses, customers, terms, usage, jurisdictions, renewals, and acceptance authority remain unresolved.

365

SECTION 365 - USAGE METERING

CODEX/03_SECTIONS/365_USAGE_METERING.txt

SHA-256 eebe930809d83391ae0a3ef4cf6778a63420742680d16dcc23d8c6ef2aea9d15

4,810 BYTES / PRESERVED SOURCE IDENTITY

SECTION 365 — Usage Metering

STATUS: CONTENT-COMPLETE

PURPOSE

Measure resource, product, service, subscription, or licensed usage accurately, transparently, securely, and with sufficient lineage for billing, capacity, entitlement, and decisions.

REQUIRED OUTCOME

A usage-metering model with event, subject, unit, source, time, aggregation, correction, privacy, completeness, idempotence, retention, consumer, and evidence.

WHY THIS MATTERS

Usage data can drive charges, access, forecasts, and capacity. Missing, duplicate, late, misclassified, or privacy-invasive measurements can create financial and customer harm.

APPLICABILITY

Apply when measured usage influences billing, licensing, subscription, capacity, product analytics, service levels, or customer decisions.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no usage, event, unit, consumer, price, entitlement, capacity, or analytic behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use data, analytics, commerce, billing, payments, licensing, subscriptions, privacy, security, observability, and records baselines.

REQUIRED INPUTS

Usage event, subject, unit, source, timestamp, context, aggregation, deduplication, correction, retention, consumer, and quality threshold.

HUMAN RESPONSIBILITIES AND AUTHORITY

Metering owners define unit and meaning; finance, product, licensing, privacy, and data owners approve use and corrections. Humans decide disputes and material backfill.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized metering metadata and bounded samples, identify anomalies, and draft reconciliation. It must not expose subject data, change events, alter billable quantity, or declare completeness without authority.

APPLICABLE EXISTING AGENT ROLES

Metering Engineer, Data Engineer, Billing Engineer, Product Analyst, Privacy Reviewer, Security Auditor, Tester, Verifier, and Financial Analyst.

REQUIRED ACTIVITIES

Define units and events; instrument; capture; timestamp; deduplicate; aggregate; validate; reconcile; correct; publish; monitor late or missing data; retain; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show subject, event, unit, source, time, aggregation, privacy, consumer, evidence, and limitation. Require human approval for correction, billing, licensing, deletion, and dispute.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use immutable event identity, idempotence, clock and late-data handling, lineage, privacy, access, correction, backfill, retention, and separation from authoritative ledger.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show usage health, completeness, duplicate, freshness, unit, consumer, evidence freshness, owner, and authorized quarantine, correct, backfill, restrict, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Metering charter; event and unit; source and lineage; quality and aggregation; correction and reconciliation; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify event, subject, unit, source, timestamp, window, method, result, quality, actor, and limitation. Count proves only its event source, window, and rules.

VERIFICATION AND ACCEPTANCE TESTS

Verify completeness, uniqueness, timing, unit, aggregation, privacy, access, late event, correction, backfill, billing, licensing, capacity, and reproducibility.

FAILURE STATES AND ANTI-PATTERNS

Duplicate charge, hidden event loss, unit drift, timestamp confusion, subject exposure, late usage ignored, correction without lineage, and meter as unquestioned truth.

RECOVERY AND RECONSIDERATION

Quarantine affected usage, preserve raw and derived versions, reconcile, correct or backfill, notify consumers, and reassess charges or access. Reopen on event, unit, source, price, or policy change.

COMPLETION CRITERIA

Events, units, subjects, sources, timing, quality, privacy, consumers, corrections, evidence, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve event and metric versions, quality, corrections, invoices, decisions, evidence, exceptions, and open gaps. Hand off to Data, Billing, Licensing, Subscription, Commerce, Analytics, Privacy, and Records.

RELATED CODEX SECTIONS

031 Data Architecture; 254 Data Testing; 358-364 Commerce, Billing, Payments, Subscriptions, Revenue, and Licensing; 366-370 Analytics and Experiments; 391-395 Data Lifecycle.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, data, commerce, finance, privacy, analytics, and evidence context, and the integration roadmap. Exact events, units, sources, subjects, quality, correction, retention, and acceptance authority remain unresolved.

366

SECTION 366 - CUSTOMER ANALYTICS

CODEX/03_SECTIONS/366_CUSTOMER_ANALYTICS.txt

SHA-256 73db18dc89083b8d8344ca1e62a99b8f7b477973ceb5ae443d57761662a8bbc5

4,816 BYTES / PRESERVED SOURCE IDENTITY

SECTION 366 — Customer Analytics

STATUS: CONTENT-COMPLETE

PURPOSE

Analyze customer behavior, needs, outcomes, support, adoption, retention, value, and risk with respect for privacy, context, uncertainty, and human interpretation.

REQUIRED OUTCOME

A customer-analytics model with purpose, population, data, consent or authority, definitions, segments, measures, cohorts, privacy, action, evidence, and limitations.

WHY THIS MATTERS

Customer data can inform service and product improvement while exposing people to surveillance, inference, unfair treatment, or misleading segmentation. Correlation and activity are not customer truth.

APPLICABILITY

Apply when customer data or feedback informs product, service, support, marketing, pricing, risk, retention, or customer-success decisions.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no customer data, segment, decision, metric, or customer-impacting use changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use data, privacy, consent, security, product, customer success, support, metrics, research, legal, and governance baselines.

REQUIRED INPUTS

Purpose, customers, data, source, authority, population, segment, metric, cohort, timeframe, action, privacy, and evidence.

HUMAN RESPONSIBILITIES AND AUTHORITY

Customer and data owners approve purpose, use, segmentation, access, retention, action, and high-impact interpretation. Humans protect customer agency and resolve ambiguity.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized aggregate or bounded data, identify patterns, and draft hypotheses. It must not infer sensitive traits, expose individuals, make customer decisions, or claim causality without authority and review.

APPLICABLE EXISTING AGENT ROLES

Customer Analyst, Data Analyst, Researcher, Product Manager, Privacy Reviewer, Statistician, User Experience Reviewer, Verifier, and Critic.

REQUIRED ACTIVITIES

Define question and purpose; inventory and minimize data; define metrics and segments; analyze; validate with customer and qualitative evidence; assess harm; recommend; monitor; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show purpose, population, data source, segment, metric, period, evidence, uncertainty, privacy, and limitation. Require human review for personal, sensitive, high-impact, and customer-facing action.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use purpose-bound data, aggregation, access, privacy, lineage, cohort and metric versioning, suppression, retention, correction, and separation of analysis from action.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show question, population, metrics, segments, data freshness, privacy, findings, owner, and authorized restrict, correct, annotate, act, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Customer-analytics charter; purpose and data; metric and segment; analysis and evidence; action and privacy; decision, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify question, population, source, data, metric, segment, period, method, result, reviewer, uncertainty, and limitation. Activity, sentiment, or retention proves only its bounded signal.

VERIFICATION AND ACCEPTANCE TESTS

Verify authority, data quality, privacy, metric, segment, cohort, bias, causality limits, customer validation, access, retention, and action. Test missing and contradictory data.

FAILURE STATES AND ANTI-PATTERNS

Surveillance by default, sensitive inference, segment as identity, correlation as cause, dashboard as customer voice, data reuse without purpose, and metric gaming.

RECOVERY AND RECONSIDERATION

Restrict or delete unauthorized data, correct analysis, notify owners, remedy affected decisions, preserve evidence, and reassess purpose and harm. Reopen on data, population, purpose, model, policy, or customer change.

COMPLETION CRITERIA

Purpose, customers, data, authority, metrics, segments, evidence, privacy, actions, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve datasets, definitions, analyses, decisions, evidence, privacy assessments, exceptions, and open gaps. Hand off to Product Analytics, Customer Success, Data, Privacy, Marketing, Support, and Governance.

RELATED CODEX SECTIONS

015 User Research; 031 Data; 218 Privacy; 336 Customer Success; 343 Product Marketing; 365 Metering; 367 Product Analytics; 369-372 Experiments and Feedback.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, customer, data, privacy, product, and evidence context, and the integration roadmap. Exact customers, purposes, sources, segments, consent, metrics, action, and acceptance authority remain unresolved.

367

SECTION 367 - PRODUCT ANALYTICS

CODEX/03_SECTIONS/367_PRODUCT_ANALYTICS.txt

SHA-256 0a4ba5a7cc6a61698395a4bef37a851a1b5a81448ed947b69f0f75c552d496c4

4,517 BYTES / PRESERVED SOURCE IDENTITY

SECTION 367 — Product Analytics

STATUS: CONTENT-COMPLETE

PURPOSE

Measure and understand product behavior, adoption, journeys, outcomes, quality, and opportunity to inform responsible product decisions.

REQUIRED OUTCOME

A product-analytics model with question, outcome, event, metric, population, cohort, funnel, experiment, data quality, privacy, decision, and limitation.

WHY THIS MATTERS

Product telemetry can show what happened but not always why, whether it helped, or who was excluded. Optimizing clicks or retention can harm comprehension, safety, or trust.

APPLICABILITY

Apply when product usage, journey, feature, quality, outcome, or experiment data informs product or operational decisions.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no product data, event, metric, journey, decision, or customer impact changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use product, user research, data, privacy, metrics, feature, experimentation, support, and governance baselines.

REQUIRED INPUTS

Question, product, user, journey, event, metric, cohort, source, period, feature, outcome, privacy, and evidence.

HUMAN RESPONSIBILITIES AND AUTHORITY

Product owners define outcome and decision; data and privacy owners approve collection and use; humans validate interpretation with research and domain judgment.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized aggregate product data, identify patterns, and draft hypotheses. It must not infer user intent, expose individuals, make product decisions, or claim causal effect without human review.

APPLICABLE EXISTING AGENT ROLES

Product Analyst, Data Analyst, Product Manager, Researcher, Statistician, Privacy Reviewer, Experimenter, Verifier, and Critic.

REQUIRED ACTIVITIES

Define question and outcome; instrument events; validate data; analyze journeys and cohorts; combine qualitative evidence; assess harm; recommend; experiment; monitor; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show question, event, metric, cohort, source, period, evidence, uncertainty, privacy, and limitation. Require human review for high-impact product, targeting, and retention decisions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use event taxonomy, identity and consent, data lineage, metric versioning, privacy, aggregation, quality, experiment separation, and feedback to product decisions.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show product question, journey, metric, cohort, freshness, data quality, findings, owner, and authorized instrument, correct, restrict, experiment, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Product-analytics charter; event and metric catalog; journey and cohort; analysis and outcome; decision, evidence, privacy, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify question, event, user or cohort, source, metric, period, method, result, reviewer, and limitation. Usage or funnel change proves neither cause nor value.

VERIFICATION AND ACCEPTANCE TESTS

Verify events, identities, consent, data quality, metrics, cohorts, journey, privacy, causality limits, experiment, and decision traceability.

FAILURE STATES AND ANTI-PATTERNS

Instrument everything, metric drift, event semantics unclear, retention as value, cohort bias, personal data exposure, and product change driven by dashboard alone.

RECOVERY AND RECONSIDERATION

Correct or withdraw data, preserve versions, notify consumers, revise product decision, and reassess user impact. Reopen on event, metric, feature, population, privacy, or product change.

COMPLETION CRITERIA

Question, outcome, events, metrics, cohorts, privacy, evidence, owners, decisions, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve event and metric versions, analyses, experiments, decisions, evidence, exceptions, and open gaps. Hand off to Product, Customer Analytics, Data, Experimentation, Feedback, and Governance.

RELATED CODEX SECTIONS

010 Product Management; 161-162 Analytics and BI; 218 Privacy; 366 Customer Analytics; 368-372 Growth, Experimentation, Optimization, and Feedback.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, product, data, privacy, analytics, and decision context, and the integration roadmap. Exact questions, events, users, metrics, consent, experiments, and acceptance authority remain unresolved.

368

SECTION 368 - GROWTH ENGINEERING

CODEX/03_SECTIONS/368_GROWTH_ENGINEERING.txt

SHA-256 70302a90fb829c697e9de3d481daf2fd7a5de7d00d62137a304990794c632a74

4,791 BYTES / PRESERVED SOURCE IDENTITY

SECTION 368 — Growth Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Design and operate responsible mechanisms for adoption, activation, retention, referral, expansion, or reach while protecting user agency, trust, accessibility, and product value.

REQUIRED OUTCOME

A growth model with objective, audience, journey, intervention, metric, experiment, privacy, safety, guardrails, feedback, and stop or rollback criteria.

WHY THIS MATTERS

Growth systems can optimize behavior rather than benefit and can create dark patterns, spam, exclusion, dependency, or harm. Growth is not inherently evidence of value.

APPLICABILITY

Apply when product, service, communication, pricing, feature, referral, onboarding, or lifecycle interventions aim to change adoption or customer behavior.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no user behavior, audience, adoption, retention, commercial, or intervention outcome changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use product, customer, analytics, privacy, trust and safety, ethics, legal, experimentation, accessibility, and commerce baselines.

REQUIRED INPUTS

Objective, audience, journey, intervention, metric, cohort, data, experiment, risk, guardrail, feedback, and stop condition.

HUMAN RESPONSIBILITIES AND AUTHORITY

Growth owners define protected outcomes and guardrails; product, privacy, legal, ethics, and user owners review interventions. Humans approve high-impact targeting and communication.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized aggregate behavior, propose hypotheses, draft experiments, and identify anomalies. It must not target sensitive traits, manipulate users, expose data, or optimize a metric without human review.

APPLICABLE EXISTING AGENT ROLES

Growth Engineer, Product Manager, Product Analyst, Researcher, Experimenter, Privacy Reviewer, Ethics Reviewer, User Experience Reviewer, Verifier, and Critic.

REQUIRED ACTIVITIES

Define value and guardrails; understand journey; design intervention; assess harm and privacy; experiment; monitor; support opt-out; stop or roll back; analyze outcome; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show objective, audience, intervention, data, metric, guardrail, evidence, uncertainty, and limitation. Require human approval for targeting, communication, pricing, consent, and stop decisions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use consent, audience controls, feature flags, experiment isolation, frequency limits, accessibility, privacy, opt-out, rollback, and harm monitoring.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show intervention, audience, metric, guardrail, complaints, evidence freshness, owner, and authorized pause, narrow, opt out, rollback, compensate, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Growth charter; audience and journey; intervention and guardrail; experiment and metric; feedback, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify objective, audience, intervention, cohort, metric, period, source, method, result, harm, and limitation. Growth metric proves only behavior change, not value or consent.

VERIFICATION AND ACCEPTANCE TESTS

Verify targeting, consent, accessibility, privacy, guardrails, experiment, harm, opt-out, communication, rollback, and user outcome. Test adverse and vulnerable cohorts.

FAILURE STATES AND ANTI-PATTERNS

Dark pattern, metric gaming, sensitive targeting, spam, coercive retention, growth as value, no opt-out, and intervention continued after harm signal.

RECOVERY AND RECONSIDERATION

Pause intervention, protect users, preserve evidence, communicate, reverse or compensate, and reassess goal and guardrails. Reopen on audience, product, policy, metric, or harm change.

COMPLETION CRITERIA

Objective, audience, intervention, metrics, guardrails, evidence, privacy, owners, stop conditions, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve interventions, cohorts, experiments, feedback, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Product, Analytics, Experimentation, Customer Success, Privacy, Ethics, and Marketing.

RELATED CODEX SECTIONS

010 Product Management; 217 Trust and Safety; 218 Privacy; 366-367 Customer and Product Analytics; 369-371 Experimentation, A/B Testing, and Optimization; 372 Feedback.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, product, analytics, privacy, ethics, trust, and decision context, and the integration roadmap. Exact objectives, audiences, interventions, guardrails, consent, metrics, and acceptance authority remain unresolved.

369

SECTION 369 - EXPERIMENTATION

CODEX/03_SECTIONS/369_EXPERIMENTATION.txt

SHA-256 7a27898ddaa9c85f32dac00a549f99296c2972ca22f437d22ede3f154d332e5d

4,893 BYTES / PRESERVED SOURCE IDENTITY

SECTION 369 — Experimentation

STATUS: CONTENT-COMPLETE

PURPOSE

Run bounded, hypothesis-driven experiments that reduce uncertainty while protecting people, data, service, fairness, safety, and decision integrity.

REQUIRED OUTCOME

An experiment model with hypothesis, objective, population, intervention, control, metric, assignment, duration, power or uncertainty, guardrail, consent or authority, analysis, stop, and decision.

WHY THIS MATTERS

An experiment can produce a precise answer to the wrong question or create harm through spillover, biased sample, peeking, interference, or unmonitored guardrail failure.

APPLICABILITY

Apply when a controlled change or comparison is used to learn causal or comparative evidence about product, process, policy, system, or user outcome.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no experiment, causal claim, intervention, population, or decision is planned and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use research, analytics, product, privacy, ethics, safety, statistics, feature, and governance baselines. Define experiment owner and stop authority.

REQUIRED INPUTS

Question, hypothesis, population, assignment, intervention, control, metric, sample, duration, guardrail, data, consent, analysis, and decision.

HUMAN RESPONSIBILITIES AND AUTHORITY

Experiment owners approve purpose, population, intervention, risk, consent, guardrails, stop, and interpretation. Humans judge ethical, practical, and customer meaning.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft hypotheses, design cases, analyze authorized aggregate results, and surface uncertainty. It must not assign sensitive participants, alter experiment, peek and conclude, expose data, or decide rollout without human review.

APPLICABLE EXISTING AGENT ROLES

Experimenter, Statistician, Researcher, Product Analyst, Product Manager, Privacy Reviewer, Ethics Reviewer, Tester, Verifier, and Critic.

REQUIRED ACTIVITIES

Define question and hypothesis; assess harm; design; assign; instrument; run; monitor guardrails; stop if needed; analyze; inspect limitations; decide; communicate; and preserve reproducibility.

HUMAN-AI INTERACTION REQUIREMENTS

Show hypothesis, population, assignment, intervention, metric, guardrail, authority, evidence, uncertainty, and limitation. Require human review for sensitive population, harm, stop, and rollout decision.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use randomization or declared alternative, isolation, exposure control, feature flag, event integrity, privacy, guardrails, stopping rule, analysis version, and rollback.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show experiment state, population, intervention, metric, guardrail, incidents, evidence freshness, owner, and authorized pause, stop, analyze, extend, rollout, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Experiment charter; hypothesis and design; population and assignment; metrics and guardrails; analysis and decision; evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify hypothesis, population, assignment, intervention, control, metric, period, method, result, uncertainty, and limitation. Statistical significance alone proves neither value nor safety.

VERIFICATION AND ACCEPTANCE TESTS

Verify assignment, exposure, contamination, sample, metric, guardrail, consent, data quality, analysis, stopping, reproducibility, and decision. Test missing data and adverse subgroup.

FAILURE STATES AND ANTI-PATTERNS

Peeking, underpowered study, biased sample, metric substitution, control contamination, harm ignored, experiment without stop, and statistically significant proxy treated as success.

RECOVERY AND RECONSIDERATION

Stop or narrow, protect users, preserve data, correct analysis, communicate, and reassess. Reopen on data, population, intervention, metric, guardrail, or external event change.

COMPLETION CRITERIA

Hypothesis, design, population, metrics, guardrails, evidence, analysis, decision, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve design, assignments, feature state, data, analysis, decisions, incidents, evidence, exceptions, and open gaps. Hand off to A/B Testing, Product Analytics, Growth, Optimization, Privacy, and Governance.

RELATED CODEX SECTIONS

001 Research; 366-368 Customer, Product, and Growth Analytics; 370 A/B Testing; 371 Optimization; 217-218 Trust and Privacy; 253 Model Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, experimentation, statistics, product, privacy, ethics, and evidence context, and the integration roadmap. Exact hypotheses, populations, metrics, power, guardrails, consent, and acceptance authority remain unresolved.

370

SECTION 370 - A OR B TESTING

CODEX/03_SECTIONS/370_A_OR_B_TESTING.txt

SHA-256 2d8f8603d3d5fc35be1acaaed6fef77e6cebf97d2e6d37692d0e41718652d161

4,615 BYTES / PRESERVED SOURCE IDENTITY

SECTION 370 — A/B Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Compare two or more controlled variants against a declared outcome using fair assignment, reliable measurement, guardrails, and accountable interpretation.

REQUIRED OUTCOME

An A/B-test model with variant, population, randomization, exposure, control, metric, sample, duration, guardrails, statistical method, stop rule, and rollout decision.

WHY THIS MATTERS

A/B testing can isolate effect but is vulnerable to biased assignment, interference, novelty, sample loss, repeated looks, and proxy optimization. A winner is not automatically a good product.

APPLICABILITY

Apply when variants of a product, interface, process, message, policy, or system are compared experimentally.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no variant comparison or causal claim is made and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use experimentation, product analytics, statistics, privacy, ethics, accessibility, feature, and governance baselines.

REQUIRED INPUTS

Variants, population, assignment, exposure, control, metric, sample, duration, guardrail, consent, analysis, and decision owner.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners approve variants, population, guardrails, stop, analysis, rollout, and interpretation. Humans review user, accessibility, safety, privacy, and ethical effect.

AI RESPONSIBILITIES AND BOUNDARIES

AI may validate authorized event data, compare variants, and draft analysis. It must not change assignment, infer causal winner, expose participants, or roll out a variant without human approval.

APPLICABLE EXISTING AGENT ROLES

A/B Test Engineer, Statistician, Product Analyst, Experimenter, Researcher, Privacy Reviewer, Accessibility Reviewer, Verifier, and Critic.

REQUIRED ACTIVITIES

Define hypothesis and variants; verify assignment; instrument; run; monitor guardrails; analyze without unsupported peeking; review segments; decide; roll out or revert; and preserve evidence.

HUMAN-AI INTERACTION REQUIREMENTS

Show variant, assignment, exposure, metric, guardrail, sample, period, evidence, uncertainty, and limitation. Require human approval for stop, winner, rollout, and subgroup action.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use stable assignment, exclusion, feature flag, exposure event, metric lineage, privacy, guardrail, holdout, rollback, and analysis reproducibility.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show test, variants, population, exposure, metric, guardrails, evidence freshness, owner, and authorized pause, stop, analyze, rollout, revert, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

A/B charter; variant and assignment; metric and guardrail; analysis and decision; rollout and rollback; evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify variant, population, assignment, exposure, metric, source, period, method, result, uncertainty, and limitation. Statistical result proves only the declared design and sample.

VERIFICATION AND ACCEPTANCE TESTS

Verify randomization, exposure, contamination, sample, metric, guardrail, privacy, accessibility, stopping, analysis, rollout, and rollback. Test assignment failure and missing event.

FAILURE STATES AND ANTI-PATTERNS

Assignment bias, sample ratio mismatch ignored, peeking, guardrail omission, winner by proxy, test pollution, and rollout without user or safety review.

RECOVERY AND RECONSIDERATION

Pause or revert, preserve assignments and data, correct analysis, notify owners, and reassess affected users. Reopen on variant, metric, population, guardrail, or product change.

COMPLETION CRITERIA

Variants, assignment, exposure, metrics, guardrails, evidence, analysis, decision, owner, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve test config, assignments, variants, results, decisions, incidents, evidence, exceptions, and open gaps. Hand off to Experimentation, Product Analytics, Growth, Optimization, Privacy, and Release.

RELATED CODEX SECTIONS

369 Experimentation; 367 Product Analytics; 368 Growth; 371 Optimization; 217-218 Trust and Privacy; 296 Feature Management; 237 Acceptance Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, experimentation, product, statistics, privacy, accessibility, and release context, and the integration roadmap. Exact variants, populations, metrics, power, guardrails, consent, and acceptance authority remain unresolved.

371

SECTION 371 - OPTIMIZATION

CODEX/03_SECTIONS/371_OPTIMIZATION.txt

SHA-256 0721ec5362fedb11155bf6e17ec0e4b3fb4ce96c4e0bc5ec4b153c86a3ed72a6

4,773 BYTES / PRESERVED SOURCE IDENTITY

SECTION 371 — Optimization

STATUS: CONTENT-COMPLETE

PURPOSE

Improve a defined outcome or constraint through measured change while protecting other required outcomes, uncertainty, safety, fairness, maintainability, and reversibility.

REQUIRED OUTCOME

An optimization model with objective, constraints, baseline, variables, method, tradeoffs, guardrails, evidence, experiment, decision, rollback, and monitoring.

WHY THIS MATTERS

Optimization can improve a proxy while worsening the actual goal or shifting cost and harm elsewhere. Local improvement is not system improvement without constraints and outcome evidence.

APPLICABILITY

Apply when a system, process, product, resource, model, or operation is deliberately tuned to improve performance, cost, quality, growth, reliability, or another outcome.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no objective, variable, tradeoff, or measurable outcome changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use requirements, metrics, experiments, performance, cost, safety, privacy, ethics, quality, reliability, and change baselines.

REQUIRED INPUTS

Objective, constraints, baseline, variables, metric, data, method, alternatives, guardrails, evidence, owner, and rollback.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners define objective and protected constraints; specialists assess domain, safety, security, and user impact; humans approve tradeoffs, deployment, and residual risk.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized data, propose candidates, simulate bounded changes, and identify tradeoffs. It must expose objective and limitations and must not alter production, optimize sensitive outcome, or accept collateral harm without human authority.

APPLICABLE EXISTING AGENT ROLES

Optimizer, Analyst, Researcher, Architect, Experimenter, Performance Engineer, User Experience Reviewer, Safety Reviewer, Verifier, and Critic.

REQUIRED ACTIVITIES

Define objective and constraints; baseline; identify variables; propose options; test or experiment; assess tradeoffs and harm; select; deploy; monitor; rollback; and review.

HUMAN-AI INTERACTION REQUIREMENTS

Show objective, constraint, baseline, metric, candidate, evidence, uncertainty, tradeoff, and limitation. Require human approval for high-impact optimization and production change.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use feature flags, experiment isolation, guardrails, rollback, multi-objective measurement, change traceability, safe bounds, and independent monitoring.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show objective, baseline, candidate, metric, constraints, harm signals, evidence freshness, owner, and authorized test, deploy, pause, rollback, constrain, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Optimization charter; objective and constraint; baseline and candidate; experiment and result; tradeoff and guardrail; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify objective, baseline, candidate, data, metric, method, period, result, tradeoff, uncertainty, and limitation. Local metric improvement proves neither global outcome nor no harm.

VERIFICATION AND ACCEPTANCE TESTS

Verify baseline, constraints, metric, candidate, experiment, guardrails, user impact, security, privacy, cost, rollback, and outcome. Test adverse and boundary conditions.

FAILURE STATES AND ANTI-PATTERNS

Proxy optimization, goal drift, local maximum, harm externalized, metric gaming, no baseline, unbounded automation, and improvement without rollback.

RECOVERY AND RECONSIDERATION

Pause or roll back, protect affected parties, preserve evidence, correct metric or objective, and reassess tradeoffs. Reopen on outcome, constraint, data, population, system, or risk change.

COMPLETION CRITERIA

Objective, constraints, baseline, variables, evidence, tradeoffs, guardrails, decision, owners, limits, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve candidates, experiments, metrics, decisions, incidents, evidence, exceptions, and open gaps. Hand off to Growth, Product, Analytics, Performance, SRE, Quality, and Operations.

RELATED CODEX SECTIONS

229 Quality Engineering; 239 Performance; 321-324 Capacity, Availability, Performance, Reliability; 366-370 Analytics and Experiments; 368 Growth; 372 Feedback.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, optimization, metrics, experiments, user, quality, and operations context, and the integration roadmap. Exact objective, constraints, baselines, data, guardrails, tradeoffs, and acceptance authority remain unresolved.

372

SECTION 372 - FEEDBACK MANAGEMENT

CODEX/03_SECTIONS/372_FEEDBACK_MANAGEMENT.txt

SHA-256 33d9ee8e987ffa0f14f41a7d5f72547a4d5c35b949270679de7ed82313f986c2

4,786 BYTES / PRESERVED SOURCE IDENTITY

SECTION 372 — Feedback Management

STATUS: CONTENT-COMPLETE

PURPOSE

Collect, protect, understand, prioritize, respond to, and learn from feedback, complaints, requests, observations, and dissent across the lifecycle.

REQUIRED OUTCOME

A feedback model with source, participant, context, consent, topic, severity, triage, owner, response, action, evidence, privacy, and closure or reopening.

WHY THIS MATTERS

Feedback is signal, not automatic truth or noise. Ignoring dissent, over-weighting loud voices, exposing reporters, or closing without response damages product and trust.

APPLICABILITY

Apply when feedback from users, customers, operators, communities, staff, suppliers, tests, incidents, or stakeholders informs decisions or improvement.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no feedback, participant, issue, decision, or improvement workflow changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use research, support, customer success, community, trust and safety, privacy, analytics, product, governance, and records baselines.

REQUIRED INPUTS

Feedback, source, participant, context, consent, issue, severity, topic, evidence, owner, response, action, and retention.

HUMAN RESPONSIBILITIES AND AUTHORITY

Feedback owners protect participants, triage fairly, respond, escalate harm, and close transparently. Humans decide priority, action, compensation, and interpretation.

AI RESPONSIBILITIES AND BOUNDARIES

AI may classify authorized feedback, summarize themes, identify contradictions, and draft response or research questions. It must not expose participants, infer intent, erase dissent, promise action, or close high-impact feedback without review.

APPLICABLE EXISTING AGENT ROLES

Researcher, User Experience Reviewer, Customer Support, Community Manager, Product Manager, Analyst, Privacy Reviewer, Trust and Safety Reviewer, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Collect and protect; authenticate or anonymize as needed; classify; triage; acknowledge; investigate; respond; route; act; communicate disposition; analyze trends; and retain or delete.

HUMAN-AI INTERACTION REQUIREMENTS

Show source, context, participant protection, theme, evidence, action, owner, and limitation. Require human review for harm, privacy, public response, policy, and closure.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Provide consent, identity or anonymity, secure intake, case and topic, deduplication, provenance, response, appeal, access, retention, and feedback-to-decision traceability.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show feedback queue, topic, severity, source, owner, response, evidence freshness, privacy, action, and authorized assign, protect, respond, escalate, close, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Feedback charter; source and participant; taxonomy and triage; response and action; trend and dissent; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify feedback, source, participant or segment, context, method, time, theme, action, result, and limitation. Volume or sentiment proves only a bounded signal.

VERIFICATION AND ACCEPTANCE TESTS

Verify intake, privacy, accessibility, classification, triage, response, escalation, action, dissent, retention, and user understanding. Test abusive, duplicate, anonymous, and conflicting feedback.

FAILURE STATES AND ANTI-PATTERNS

Feedback theater, reporter exposure, loudest voice, sentiment as truth, no response, closure by categorization, and feedback collected without action owner.

RECOVERY AND RECONSIDERATION

Protect participant, correct record, respond, escalate, preserve evidence, remedy harm, and reassess decision. Reopen on new feedback, harm, population, product, or policy change.

COMPLETION CRITERIA

Source, participant, context, privacy, taxonomy, triage, owner, response, evidence, action, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve feedback, participant protections, responses, actions, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Product, Research, Support, Community, Analytics, Privacy, and Governance.

RELATED CODEX SECTIONS

001 Research; 334-337 Support, Success, and Community; 366-371 Analytics, Growth, Experimentation, and Optimization; 217-218 Trust and Privacy; 225 Records.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, feedback, research, customer, privacy, product, and evidence context, and the integration roadmap. Exact sources, participants, taxonomy, staffing, response, retention, and acceptance authority remain unresolved.

373

SECTION 373 - ROADMAP MANAGEMENT

CODEX/03_SECTIONS/373_ROADMAP_MANAGEMENT.txt

SHA-256 96ea2d8d0113db13e109970c29b5db3a2f64f1023c278150db5eef9ad4426b91

4,837 BYTES / PRESERVED SOURCE IDENTITY

SECTION 373 — Roadmap Management

STATUS: CONTENT-COMPLETE

PURPOSE

Sequence outcomes, bets, capabilities, dependencies, options, commitments, and learning over time while preserving strategy, evidence, capacity, and change authority.

REQUIRED OUTCOME

A roadmap model with objective, horizon, themes, initiatives, outcomes, evidence, dependencies, capacity, risk, decision gates, confidence, ownership, and change history.

WHY THIS MATTERS

Roadmaps influence investment and expectation but are not promises by default. False precision, feature lists without outcomes, and stale priorities can consume capacity while avoiding hard choices.

APPLICABILITY

Apply when work is sequenced, prioritized, forecast, committed, funded, deferred, or retired across a portfolio, product, program, or capability.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no priority, sequence, investment, dependency, outcome, or commitment changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use mission, strategy, product, portfolio, research, market, finance, capacity, risk, architecture, feedback, and delivery baselines.

REQUIRED INPUTS

Objectives, outcomes, initiatives, evidence, dependencies, capacity, cost, risks, options, stakeholders, constraints, decisions, and review cadence.

HUMAN RESPONSIBILITIES AND AUTHORITY

Roadmap owners facilitate priorities; accountable humans choose tradeoffs, investment, commitment, de-scoping, and stop criteria. Stakeholders can challenge assumptions and evidence.

AI RESPONSIBILITIES AND BOUNDARIES

AI may summarize evidence, map dependencies, draft options, and identify inconsistency. It must not set priority, promise delivery, invent confidence, or remove work without accountable human decision.

APPLICABLE EXISTING AGENT ROLES

Planner, Product Manager, Portfolio Manager, Analyst, Researcher, Architect, Financial Analyst, Critic, Verifier, and Coordinator.

REQUIRED ACTIVITIES

Define objective and horizon; collect evidence; frame outcomes; identify options and dependencies; assess capacity and risk; prioritize; publish confidence and assumptions; review; change; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show objective, outcome, evidence, confidence, dependency, capacity, cost, risk, owner, and limitation. Require human approval for priority, commitment, funding, cancellation, and public roadmap claim.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link roadmap item to outcome, requirements, architecture, teams, resources, dependencies, evidence, release, and retirement. Preserve option and decision history.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show roadmap horizon, outcomes, priorities, capacity, dependencies, risks, evidence freshness, owner, confidence, and authorized prioritize, defer, fund, stop, revise, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Roadmap charter; outcome and initiative; evidence and confidence; dependency and capacity; decision, commitment, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify objective, item, outcome, source, date, assumption, confidence, capacity, cost, risk, decision, and limitation. A roadmap position proves intent, not delivery or value.

VERIFICATION AND ACCEPTANCE TESTS

Verify outcome, evidence, dependency, capacity, cost, risk, owner, stakeholder understanding, decision gate, confidence, and change traceability.

FAILURE STATES AND ANTI-PATTERNS

Feature list as strategy, false dates, priority by loudness, capacity ignored, dependency hidden, confidence invented, sunk-cost continuation, and roadmap as promise without authority.

RECOVERY AND RECONSIDERATION

Mark stale, re-prioritize, communicate, preserve decisions, stop or de-scope, and reassess outcomes and capacity. Reopen on mission, evidence, market, strategy, dependency, or risk change.

COMPLETION CRITERIA

Objectives, outcomes, items, evidence, confidence, capacity, cost, dependencies, owners, decisions, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve roadmap versions, decisions, evidence, commitments, assumptions, changes, exceptions, and open gaps. Hand off to Product, Portfolio, Program, Architecture, Finance, Engineering, and Governance.

RELATED CODEX SECTIONS

004 Evidence and Truth; 005 Mission and Strategy; 006 Product Strategy; 007 Portfolio; 010 Product Management; 018 Risk; 366-373 Analytics, Growth, Feedback, and Roadmap.

CODEX/03_SECTIONS/373_ROADMAP_MANAGEMENT.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, strategy, product, portfolio, finance, evidence, and capacity context, and the integration roadmap. Exact objectives, stakeholders, capacity, funding, confidence, dates, and acceptance authority remain unresolved.